



**VIDYA VIKAS INSTITUTE OF
ENGINEERING AND TECHNOLOGY,
MYSORE**

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VTU Previous Question Papers

Electronics & Communication Engineering”

III, V & VI Semester

Dec/Jan-2024

VVIET

Bannur road, Alanahalli, Mysore-570028

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Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Digital System Design using Verilog

Digital System Design using Verilog

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. *M* : Marks , *L*: Bloom's level , *C*: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Design a combinational logic truth table so that an output is generated indicating when a majority of four inputs is true.	4	L3	CO1
	b.	Find the prime implicants and the essential prime implicants of the following Boolean functions using Karnaugh maps. i) $f(a, b, c, d) = \Sigma(1, 5, 6, 7, 11, 12, 13, 15)$ ii) $f(a, b, c, d) = \Sigma(0, 1, 4, 5, 9, 11, 13, 15)$	8	L4	CO1
	c.	Simplify the given boolean function using Quine McCluskey minimization technique for the function $O = f(a, b, c, d) = \Sigma(0, 1, 2, 3, 6, 7, 8, 9, 14, 15)$	8	L3	CO1
OR					
Q.2	a.	Place the following equations into the proper canonical form: i) $P = f(a, b, c) = ab' + ac' + bc$ ii) $G = f(w, x, y, z) = w'x + yz'$	4	L3	CO1
	b.	Find the minimal sum and minimal product for the following Boolean functions using Karnaugh maps i) $f(a, b, c, d) = \bar{a}\bar{b}d + bcd + a\bar{b}d + \bar{b}cd$ ii) $f(a, b, c, d) = (a + \bar{b})(a + c + d)(\bar{a} + \bar{b} + d)(a + \bar{c} + d)$	8	L4	CO1
	c.	Simplify the given boolean function using quine. McCluskey minimization technique for the function. $s = f(a, b, c, d) = \Sigma(1, 3, 13, 15) + \Sigma d(8, 9, 10, 11)$	8	L3	CO1
Module – 2					
Q.3	a.	Design and explain binary full adder with block diagram, Karnaugh map and logic circuit.	10	L3	CO2
	b.	Define decoder, write the symbol, truth table and logic circuit for 3:8 line decoder using minterm generator.	10	L2	CO2
OR					
Q.4	a.	Define multiplexer, write the symbol, truth table and logic circuit for 4:1 multiplexer using enable input.	10	L2	CO2
	b.	Realize the Boolean function $f(w, x, y, z) = \Sigma(0, 1, 5, 6, 7, 9, 12, 15)$ i) Using 8:1 MUX ii) Using 4:1 MUX	10	L2	CO2

1 of 2

Module – 3

Q.5	a.	Develop the characteristic equation for i) SR flip flop ii) JK flip flop iii) D flip flop iv) T flip flop.	10	L3	CO3
	b.	Explain serial in, parallel at unidirectional shift register and parallel in serious out unidirectional shift register.	10	L2	CO3

OR

Q.6	a.	Explain Mod-4 ring counter and Mod-8 twisted ring counter with logic diagram and counting sequence.	10	L2	CO3
	b.	Design a synchronous Mod-6 counter using clocked D-flip flop.	10	L3	CO3

Module – 4

Q.7	a.	Explain logical operators and relational operators used in verilog.	8	L2	CO4
	b.	Illustrate i) NETS ii) Register iii) Vector iv) integer data types with an example.	8	L2	CO4
	c.	Write a verilog code for full adder using data flow description style.	4	L2	CO4

OR

Q.8	a.	Illustrate the structure of behavioural description with an example using half adder.	8	L2	CO4
	b.	Illustrate the structure of verilog module with an example using half subtractor.	8	L2	CO4
	c.	Write a verilog code for binary to gray using behavioural description style.	4	L2	CO4

Module – 5

Q.9	a.	Write the syntax of IF and EISE-IF with an example.	8	L2	CO4
	b.	Write logic symbol, flowchart and program for D-latch using behavioural description style.	8	L2	CO4
	c.	Write a verilog code for 8:1 MUX using behavioural description style.	4	L2	CO4

OR

Q.10	a.	Explain the structure of structural model with built in gates using example of half adder. Also mention an primitive built in gates.	8	L2	CO4
	b.	Write a verilog code of a 3-bit ripple carry adder using structural description model.	8	L2	CO4
	c.	Write a verilog code of SR flip flop using behavioural description style.	4	L2	CO4

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Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Electronic Principles and Circuits

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Derive expressions V_{in} , V_{out} and A_V for a common emitter circuit with ac equivalent circuit with π – model.	12	L1	CO2
	b.	What is the voltage gain and output voltage across the load resistor of V_{DB} amplifier? $R_1 = 10\text{ k}\Omega$, $R_2 = 2.2\text{ k}\Omega$, $R_C = 3.6\text{ k}\Omega$, $R_E = 1\text{ k}\Omega$, $R_L = 10\text{ k}\Omega$, $V_{CC} = 10\text{ V}$, $V_{BE} = 0.7\text{ V}$ and $V_{in} = 2\text{ mV}$.	8	L1	CO1
OR					
Q.2	a.	With a neat diagram, explain loading effect of input impedance.	10	L1	CO1
	b.	Explain three types of Bias circuit, (i) Emitter feedback bias. (ii) Collector feedback bias and (iii) Collector and emitter feedback.	10	L1	CO1
Module – 2					
Q.3	a.	Explain the three biasing methods to bias MOS amplifiers with neat circuit diagram.	10	L2	CO2
	b.	Explain the T-equivalent circuit model of MOSFET.	10	L3	CO2
OR					
Q.4	a.	With a small signal equivalent model of MOSFET, derive an expression of voltage gain and transconductance.	10	L2	CO2
	b.	Explain common source follower and derive the expression of voltage gain with necessary equation.	10	L2	CO2
Module – 3					
Q.5	a.	Explain R and 2R resistor Digital to Analog converter and also derive the expression of output voltage.	10	L2	CO3
	b.	With a neat circuit diagram, explain the operation of Monostable multivibrator.	10	L2	CO3
OR					
Q.6	a.	With a neat diagram, explain operation of RC-phase shift oscillator using op-amp. Write the expression for frequency of oscillations.	8	L2	CO3
	b.	With a net diagram, explain operation of crystal oscillator using BJT and Write necessary equations.	6	L2	CO3
	c.	A crystal has these values $L = 3\text{ H}$, $C_s = 0.05\text{ PF}$, $R = 2\text{ k}\Omega$ and $C_m = 10\text{ PF}$. What are the series and parallel resonant frequencies of the crystal?	6	L3	CO3

Module – 4					
Q.7	a.	Explain the first order Low Pass filter with frequency response.	10	L2	CO4
	b.	Explain the two types of Band Pass filters.	10	L2	CO4
OR					
Q.8	a.	Explain the four types of Negative feedback circuits.	10	L2	CO4
	b.	Explain the working of 2 nd order high pass filter with a neat circuit and frequency response.	10	L2	CO4
Module – 5					
Q.9	a.	Explain two load lines with necessary circuit diagram and equations.	10	L2	CO5
	b.	With a neat diagram, explain the working of a Thyristor.	10	L2	CO5
OR					
Q.10	a.	Explain Basic Construction and working of IGBTs with necessary figure.	10	L2	CO5
	b.	With a neat diagram, explain the working of UJT relaxation oscillator.	10	L2	CO5

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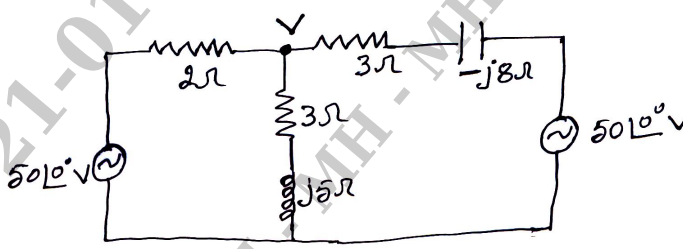
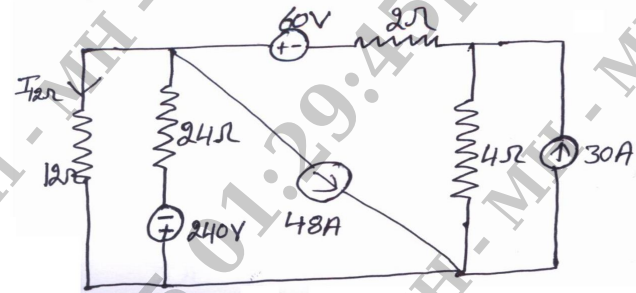
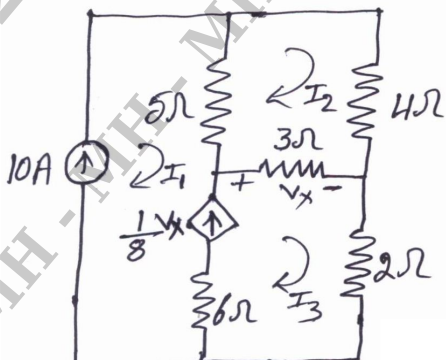
Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Network Analysis

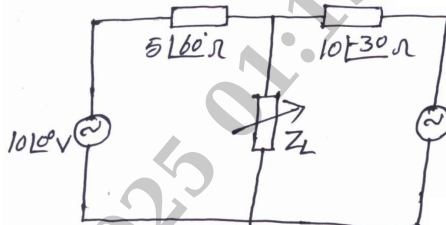
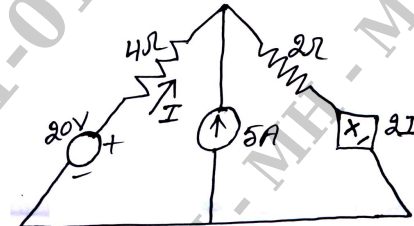
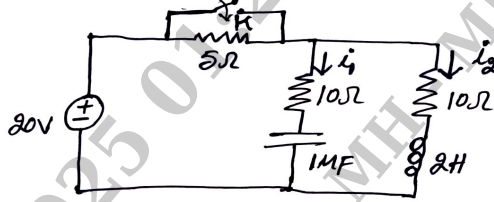
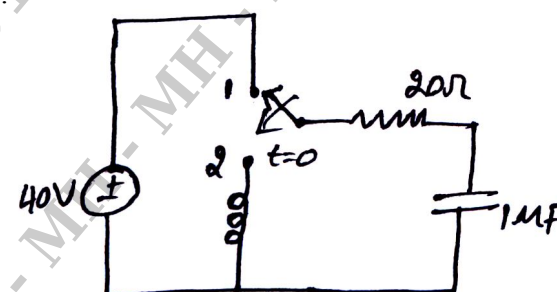
Time: 3 hrs.

Max. Marks: 100

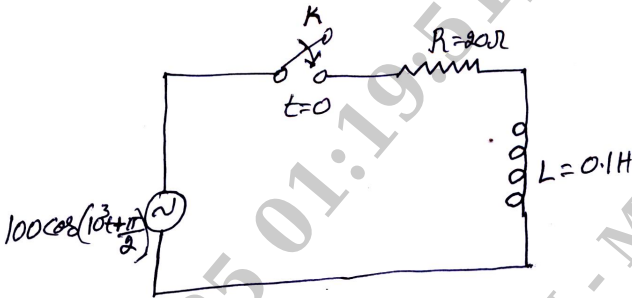
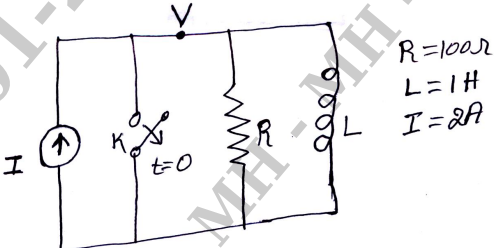
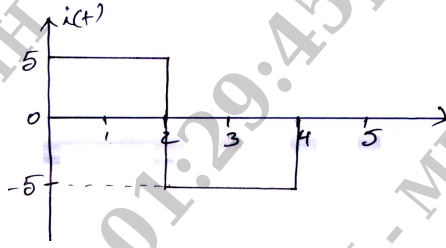
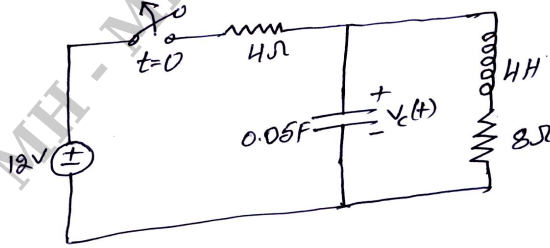
Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
 2. M : Marks , L: Bloom's level , C: Course outcomes.

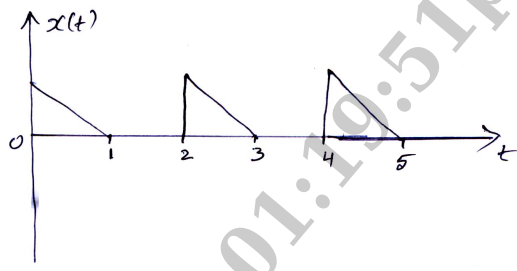
Module – 1			M	L	C
Q.1	a.	Three impedances are connected in Delta. Obtain the star equivalent of the network.	7	L3	CO1
	b.	For the circuit shown in Fig. Q1(b). Find the voltage 'V' at node by using nodal analysis. 	6	L3	CO1
	c.	Determine the current in 12Ω resistor shown in Fig. Q1(c) using source transformation method. 	7	L3	CO1
OR					
Q.2	a.	Find the loop currents I_1 , I_2 , and I_3 in the circuit shown in Fig. Q2(a). 	7	L3	CO1

	b. Determine the resistance between the terminals X, Y using star delta transformation in the network shown in Fig. Q2(b). Fig. Q2(b)	6	L3	CO1
	c. Use the nodal analysis to find the value of V_X and the circuit shown in Fig. Q2(c). Such that the current through $(2 + j3) \Omega$ Impedance is zero. Fig. Q2(c)	7	L3	CO1
Module – 2				
Q.3	a. State and prove Superposition theorem.	7	L2	CO2
	b. For the circuit shown in Fig. Q3(b), obtain the Thevenin's equivalent circuit. Fig. Q3(b)	7	L3	CO2
	c. Using Millman's theorem, find current flowing through $(3 + j4) \Omega$ impedance for the circuit shown in Fig. Q3(c). Fig. Q3(c)	6	L3	CO2
OR				

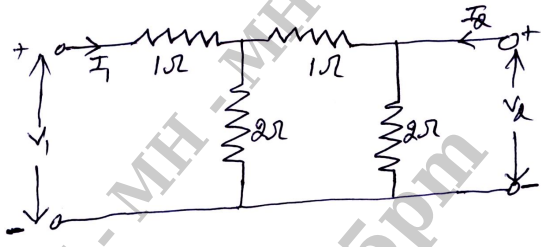
Q.4	a.	State and prove Norton's theorem.	7	L2	CO2
	b.	Find the value of Z_L for Maximum Power transfer and the value of Maximum power for the circuit shown in Fig. Q4(b).	6	L3	CO2
		Fig. Q4(b) 			
	c.	Find current 'I' using Super position theorem for the circuit shown in Fig. Q4(c).	7	L3	CO2
		Fig. Q4(c) 			
Module – 3					
Q.5	a.	Use the concepts of initial condition to illustrate the voltage behavior in inductor circuit for DC supply.	6	L3	CO3
	b.	In the circuit steady state is reached with switch 'K' open. The switch is closed at $t = 0$. Compute i , di/dt and d^2i/dt^2 at $t = 0^+$.	7	L3	CO3
		Fig. Q5(b) 			
	c.	The switch is moved from position (1) to position (2) at $t = 0$. The steady state has been reached before switching. Computer i , di/dt and d^2i/dt^2 at $t = 0^+$ for Fig. Q5(c).	7	L4	CO3
		Fig. Q5(c) 			

OR

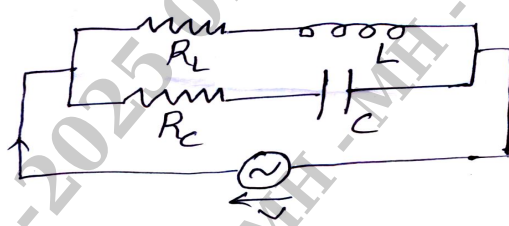
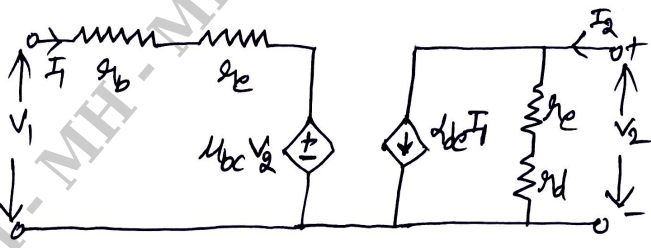
Q.6	a. In the circuit shown in Fig. Q6(a), determine complete solution for current when switch 'K' is closed at $t = 0$. Fig. Q6(a) 	10	L3	CO3
	b. Compute v, dv/dt, d^2v/dt^2 at $t = 0^+$ for the circuit shown in below Fig. Q6(b), when the switch K is opened at $t = 0$. Fig. Q6(b) 	10	L4	CO3
Module – 4				
Q.7	a. Using waveform synthesis method to express the voltage pulse in terms of unit step. Find i) $L\{i(t)\}$ ii) $L\{\int i(t).dt\}$. Fig. Q7(a) 	8	L3	CO4
	b. State and prove initial value and final value theorem for Laplace transform.	6	L2	CO4
	c. Obtain the Laplace transform of step and ramp function with relevant expressions.	6	L3	CO4
OR				
Q.8	a. Determine $i_L(t)$ for $t \geq 0$ using Laplace transform for circuit shown in Fig. Q8(a). Fig. Q8(a) 	10	L3	CO4

	b. Find the Laplace transform of the periodic signal $x(t)$ as shown in Fig. Q8(b).	10	L3	CO4
	 Fig. Q8(b)			

Module – 5

Q.9	a. Define Z – parameters. Determine Y parameters in terms of Z – parameters.	6	L3	CO5
	b. Show that resonant frequency is geometric mean of cut off frequency in series R – L – C circuit.	7	L3	CO5
	c. Apply the two – port network analysis technique to determine ABCD – parameters of the network shown in Fig. Q9(c).	7	L3	CO5
	 Fig. Q9(c)			

OR

Q.10	a. Derive the expression for the resonant frequency of the circuit shown in Fig. Q10(a). Also show that the circuit resonates at all frequency if $R_L = R_C = \sqrt{\frac{L}{C}}$	10	L3	CO5
	 Fig. Q10(a)			
	b. The model of a transistor in the CE mode is shown in Fig. Q10(b). Determine the h – parameters.	10	L3	CO5
	 Fig. Q10(b)			

CBCS SCHEME

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BEC501

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Technological Innovation and Management

Entrepreneurship

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Explain the different roles played by Managers.	10	L1	CO1
	b.	Describe the managerial skills required using skill-mix diagram.	10	L1	CO1
OR					
Q.2	a.	Explain the various steps involved in planning.	10	L2	CO1
	b.	Explain all the steps in rational decision making with a neat diagram.	10	L2	CO1
Module – 2					
Q.3	a.	Define organization. Briefly explain the principles of organizing.	10	L2	CO2
	b.	What is recruitment? Explain the steps in the selection process.	10	L2	CO2
OR					
Q.4	a.	Explain Maslow's need hierarchy theory with a neat diagram along with examples.	10	L2	CO2
	b.	Discuss Autocratic, Democratic and Free-rein leadership styles.	10	L2	CO2
Module – 3					
Q.5	a.	Define Social Audit. Explain the benefits and limitations of social audit.	10	L2	CO3
	b.	Explain the different views on social responsibility of business.	10	L2	CO3
OR					
Q.6	a.	Explain different types of entrepreneurs by defining an entrepreneur.	10	L2	CO3
	b.	Explain entrepreneurial development cycle.	10	L2	CO3
Module – 4					
Q.7	a.	Explain the different Government policy and development of the small scale sector in India.	10	L2	CO4
	b.	Explain the problems for small scale industries.	10	L2	CO4
OR					
Q.8	a.	Explain the identification of business opportunities in India.	10	L2	CO4
	b.	Explain in detail the project feasibilities.	10	L2	CO4
Module – 5					
Q.9	a.	What are the reasons for failure of some business plans?	10	L2	CO4
	b.	Explain the Government schemes for funding business.	10	L2	CO4
OR					
Q.10	a.	Explain the challenges and difficulties in starting an enterprise.	10	L2	CO4
	b.	Describe the limitations and differences of PERT and CPM.	10	L2	CO4

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CBCS SCHEME

USN

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BEC502

Fifth Semester B.E/B.Tech. Degree Examination, Dec.2024/Jan.2025 Digital Signal Processing

Time: 3 hrs.

Max. Marks:100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
1	a.	List and discuss different discrete time signals.	7	L2	CO1
	b.	Explain the steps of converting analog to digital signal in terms of frequencies.	7	L2	CO1
	c.	Discuss the advantages and limitations of Digital Signal Processing (DSP).	6	L2	CO1
OR					
2	a.	With an example, explain how to verify any signal is periodic or Not.	6	L2	CO1
	b.	Derive the equation for output of a LTI system and list the steps of convolution.	8	L3	CO2
	c.	Write a program to generate : i) Circuit step sequence ii) Sinusoidal sequence.	6	L3	CO2
Module – 2					
3	a.	Describe the properties of Z – transformation.	7	L3	CO2
	b.	Show that Discrete Fourier Transform (DFT) is a Linear Transformation.	7	L3	CO2
	c.	Compute the 4-point DFT of $x(n) = \{1, 1, 0, 0\}$.	6	L3	CO2
OR					
4	a.	Compute the N-point DFT of, $x(n) = e^{j\omega_0 n}$.	6	L3	CO2
	b.	State and prove symmetry property of DFT for real valued sequence.	6	L3	CO2
	c.	Compute circular convolution of sequences : $x_1(n) = \{2, 1, 2, 1\}$ and $x_2(n) = \{1, 2, 3, 4\}$.	8	L3	CO2
Module – 3					
5	a.	State and prove circular time shift property of DFT.	6	L3	CO2
	b.	Compare DFT and FFT with examples.	6	L2	CO3
	c.	Compute Radix – 2 DIT FFT of the following – sequence, $x(n) = n + 1$, for $0 \leq n \leq 7$.	8	L3	CO3
OR					
6	a.	State and prove Parseval's theorem for – DFT's.	6	L3	CO2
	b.	Explain overlap – save method used for the convolution of long input sequences.	6	L2	CO3
	c.	Develop an algorithm for Radix – 2 FFT without using built in function.	8	L3	CO3
1 of 2					

Module – 4

7	a.	Obtain the frequency response expression for the symmetric linear phase FIR filter.	8	L3	CO4
	b.	Compare different windows used to design FIR filters.	6	L2	CO4
	c.	Design an FIR filter using hamming window for $N = 7$. The desired frequency response is given by $H_d(\omega) = \begin{cases} e^{-j3\omega} & \omega \leq \frac{3\pi}{4} \\ 0, & \frac{3\pi}{4} < \omega \leq \pi \end{cases}$	6	L3	CO4

OR

8	a.	Discuss the characteristics of practical frequency selective filters.	6	L3	CO4
	b.	Explain the steps of designing linear phase FIR high pass filter.	8	L2	CO4
	c.	Realize the system function of following FIR filter in cascade form. $H(z) = 1 - 2z^{-1} + \frac{1}{2}z^{-2} + \frac{1}{2}z^{-3} - \frac{1}{2}z^{-4}$.	6	L3	CO4

Module – 5

9	a.	Explain the design procedure of analog Butter worth lowpass prototype – filter?	8	L3	CO5
	b.	Construct the system function in S – domain for $N = A$.	6	L3	CO5
	c.	Realize direct form – II for the IIR filter represented by $y(n) - \frac{1}{4}y(n-1) + \frac{1}{8}y(n-2) = x(n) + \frac{1}{2}x(n-2)$.	6	L3	CO5

OR

10	a.	Design the digital IIR filter for following details. –3dB gain at 0.5π rads and the stop band attenuation of 15dB at 0.75π rads. Assume $T_s = 15$.	8	L3	CO5
	b.	Explain the significance of: i) Prewarping ii) Bilinear transformation.	6	L2	CO5
	c.	Obtain the direct form-I realization of following IIR filter : $H(z) = \frac{1 + 0.4z^{-1}}{1 - 0.5z^{-1} + 0.06z^{-2}}$	6	L3	CO5

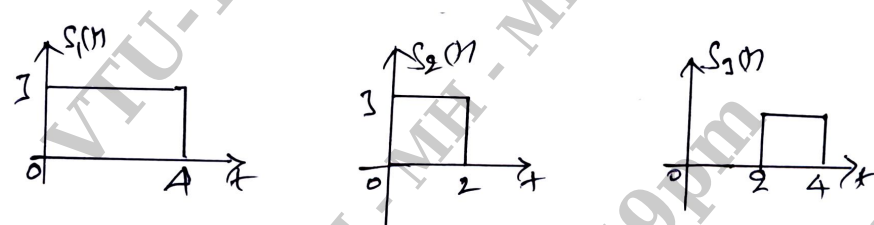
Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Digital Communication

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1				M	L	C
Q.1	a.	Explain Hilbert transform and its properties.		6	L2	CO1
	b.	Describe the canonical representation of bandpass signal.		7	L2	CO1
	c.	Describe the correlation receiver with neat diagram.		7	L2	CO1
OR						
Q.2	a.	Apply gram Schmidt orthogonalization procedure find the set of orthonormal basis function to represent the signals $S_1(t)$, $S_2(t)$ and $S_3(t)$ as shown in Fig.Q2(a). Also express each of these figures in terms of set of basis function.		10	L3	CO1
		 Fig.Q2(a)				
	b.	Derive the equation for converting continuous AWGN channel into a vector channel.		10	L2	CO1
Module – 2						
Q.3	a.	Describe with a neat diagram, the generation and detection of BPSK signal.		8	L2	CO2
	b.	Define bandwidth efficiency. Tabulate the comment on the bandwidth efficiency of M-ary PSK signal.		8	L2	CO2
	c.	Encode the binary sequence using DPSK 11011011. Assume reference bit as 1.		4	L2	CO2
OR						
Q.4	a.	Derive the expression for probability of error of QPSK signal.		8	L2	CO2
	b.	Discuss the non-coherent detection of BFSK signal.		8	L2	CO2
	c.	Calculate the average power required for a DPSK signal operation at a data rate of 1000 bit/sec, over a band-pass channel having a bandwidth of 3000 Hz, $\frac{N_0}{2} = 10^{-10}$ W/Hz probability of error $P_e = 10^{-5}$.		4	L3	CO2
Module – 3						
Q.5	a.	Define entropy and summarize its properties.		6	L2	CO3
	b.	A source has five symbols $S = \{S_1, S_2, S_3, S_4, S_5\}$ with probabilities $P = \{0.4, 0.2, 0.2, 0.1, 0.1\}$ respectively. compute the source code using Huffman binary coding. Also find the average length and entropy.		8	L3	CO3
	c.	Briefly discuss instantaneous code with an example.		6	L2	CO3
OR						
Q.6	a.	Derive the expression for mutual information and summarize its properties.		10	L2	CO3
	b.	Derive the expression for the channel capacity of binary symmetric channel.		10	L3	CO3

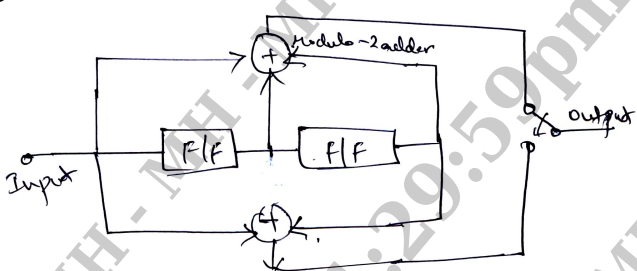
Module – 4

Q.7	a.	Indicate the advantages and disadvantages of error control coding. Also differentiate between block code and convolution code.	8	L2	CO4
	b.	If 'C' is a valid code vector then show that $CH^T = 0$ where H is parity check matrix of code.	5	L2	CO4
	c.	Design an encoder for the (7, 4) binary cyclic code generated by : $g(x) = 1 + x + x^3$ for the message vector [1001].	7	L3	CO4

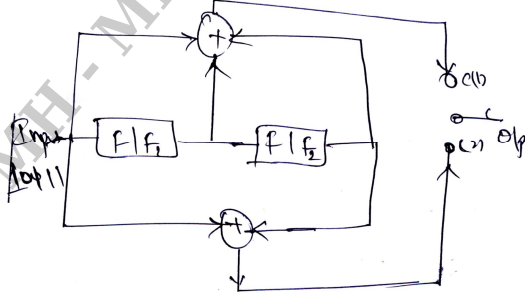
OR

Q.8	a.	Describe the block diagram of generator and parity check matrix with equation. Also write the syndrome equation and list its properties.	10	L2	CO4
	b.	A (7, 4) Linear block code has : $P = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$ i) All possible code vector ii) Determine the Hamming weight of each code word iii) If the received vector is [1100010]. Determine its syndrome correct the codeword.	10	L3	CO4

Module – 5

Q.9	a.	For a given convolutional encoder shown in Fig.Q9(a), with D = 10011. Compute output sequence using transform domain approach. Also draw the code free diagram.	10	L3	CO5
		 Fig.Q9(a)	10	L3	CO5
	b.	Describe the recursive systematic convolutional code encoder with an example.	10	L3	CO5

OR

Q.10	a.	A convolution encoder has two flip-flop with two states, three modulo – 2 adders and an output multiplexer. The generator sequences of the encoder. $g^{(1)} = (1, 0, 1)$, $g^{(2)} = (1, 1, 0)$, $g^{(3)} = (1, 1, 1)$. i) Generator matrix [G] ii) Draw the encoder block diagram iii) Calculate the codeword for the message input vector 11101.	10	L3	CO5
	b.	For a given convolution encoder shown in Fig.Q10(b). Build state table, state transaction table, sketch diagram and describe the Trellis diagram for the input message vector (10111).	10	L3	CO5
		 Fig.Q10(b)			

CBCS SCHEME

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BEC515A

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Intelligent Systems and Machine Learning Algorithms

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Explain the history of AI.	10	L2	CO1
	b.	Define AI. Explain the foundation of AI in detail.	10	L2	CO1
OR					
Q.2	a.	Explain properties of task environment.	10	L2	CO1
	b.	Differentiate between simplex reflex agents and model based reflex agents.	10	L2	CO1
Module – 2					
Q.3	a.	List and explain the components required to define a problem.	10	L2	CO2
	b.	Explain goal formulation and problem formulation with examples.	10	L2	CO2
OR					
Q.4	a.	Explain Breadth First Search and Depth First Search algorithms with an example.	10	L3	CO2
	b.	Illustrate different methods of evaluating an algorithm's performance.	10	L2	CO2
Module – 3					
Q.5	a.	Describe greedy best first search as an informed search strategy.	10	L2	CO2
	b.	Explain knowledge based agent with a generic knowledge based agent program.	10	L2	CO3
OR					
Q.6	a.	Describe syntax and semantics with respect to propositional logic.	10	L2	CO3
	b.	Explain Wumpus World with respect to artificial intelligence.	10	L2	CO3
Module – 4					
Q.7	a.	What is Machine Learning? Explain with specific examples.	06	L2	CO4
	b.	Explain perspectives and issues in Machine Learning.	04	L2	CO4
	c.	Explain types of Machine Learning System.	10	L2	CO4
OR					
Q.8	a.	Describe the main challenges of Machine Learning.	10	L2	CO4
	b.	Explain: (i) Find S algorithm (ii) Candidate elimination algorithm	10	L2	CO4
Module – 5					
Q.9	a.	Explain working with Real data and Get the data.	10	L2	CO5
	b.	Write a note on Launch, Monitor and Maintain your system.	10	L2	CO5
OR					
Q.10	a.	Describe the steps involved in preparing the data for machine learning model.	10	L2	CO5
	b.	Explain MNIST with respect to Machine Learning.	10	L2	CO5

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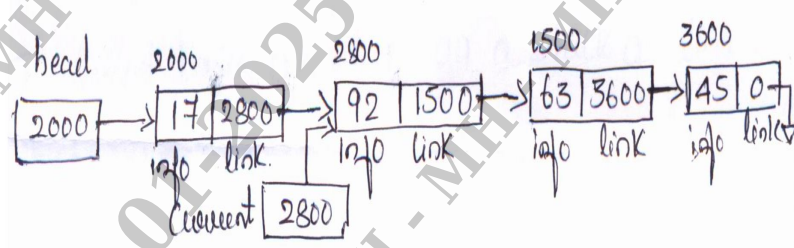
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Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Data Structures using C++

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1				M	L	C
Q.1	a.	Explain four phases of software development process.		8	L1	CO1
	b.	Explain the following with suitable example, (i) new operator (ii) delete operator (iii) friend function (iv) virtual function		8	L2	CO1
	c.	Explain the properties of constructors and destructors.		4	L2	CO1
OR						
Q.2	a.	Develop a C++ program to define base class rectangleType and the members to implement the properties of a rectangle. (i) Set Dimension (ii) Get length (iii) Get width (iv) Area (v) Perimeter (vi) Print (vii) Default constructor and constructor with parameters of the baseclass. And also define class BOXType which is delivered from the class rectangleType and implement the properties of a box. (i) Set Dimension (ii) Get Length (iii) Get Width (iv) Get Height (v) Area (vi) Volume (vii) Print (viii) Default constructor and constructor with parameters of derived class.		12	L1	CO1
	b.	Explain class inheritance as, (i) Public inheritance (ii) Private inheritance (iii) Protected inheritance		8	L1	CO1
Module – 2						
Q.3	a.	Consider the linked list shown in Fig. Q3 (a). Assume that the nodes are in the usual info-link form. Find the output of each of the following C++ statements.		5	L1	CO2
 Fig. Q3 (a) (i) Current (ii) Current → info (iii) Current → link (iv) Current → link → info (v) Head → link → link (vi) Head → link → link → info (vii) Head → link → link → link (viii) Current → link → link → link (ix) Current → link → link → link → info						

	b.	Define a class stackADT, which performs basic operations of stack: (i) Initialize stack (ii) isEmptyStack (iii) isFullStack (iv) push (v) top (vi) pop	10	L2	CO2
	c.	Write difference between arrays and linked lists.	5	L2	CO2
OR					
Q.4	a.	(i) Define a class orderedLinkedList, which specifies the members to implement the basic properties of linked list. (ii) Draw the UML class diagram, of the class orderedLinkedList and inheritance hierarchy.	6	L1	CO2
	b.	Write a C++ code, to insert an item in an ordered linked list.	6	L2	CO2
	c.	Implement basic operation on a stack as linked list.	8	L3	CO2
Module – 3					
Q.5	a.	Define a queue and explain various queue operations.	7	L1	CO3
	b.	Explain the working of binary search algorithm.	8	L1	CO3
	c.	Define the following functions for queue, (i) Initialize queue. (ii) Delete queue	5	L2	CO3
OR					
Q.6	a.	Define the following hash functions : (i) Mid-square (ii) Folding (iii) Division	6	L1	CO3
	b.	Explain various collision resolution techniques.	10	L1	CO3
	c.	Write a C++ program to implement selection sort.	4	L2	CO3
Module – 4					
Q.7	a.	Define : (i) Binary tree (ii) Use struct to define a node of a binary trees. (iii) Level of a node. (iv) Height of a binary tree.	6	L1	CO4
	b.	Explain three different binary tree traversals in detail.	10	L2	CO4
	c.	Write an algorithm for non-recursive post order traversal.	4	L3	CO4
OR					
Q.8	a.	Define a class binarysearchtree, and implement the following operations : (i) Insert (ii) Delete	10	L1	CO4
	b.	What is a B-tree of order m and explain its properties and explain an algorithm to insert an item in a B-tree.	10	L3	CO4
Module – 5					
Q.9	a.	Explain how to represent a graph in computer memory.	10	L1	CO5
	b.	Explain shortest path algorithm in detail.	10	L1	CO5
OR					
Q.10	a.	Explain Prim's minimum spanning tree algorithm.	10	L2	CO5
	b.	Explain Eulers circuits.	4	L2	CO4
	c.	Explain Breadth first topological ordering algorithm.	6	L2	CO4

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BEC515D

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Satellite and Optical Communication

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Explain the Kepler law of planetary motion. Also derive the expression for orbital period.	10	L2	CO1
	b.	A satellite is orbiting earth in a uniform circular orbit at a height of 630 km from the surface of earth. Assuming the radius of earth and its mass to be 6370 km and 5.98×10^{24} kg respectively. Determine the velocity of the satellite. (Take gravitational const $G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$).	10	L3	CO1
OR					
Q.2	a.	The apogee and perigee distance of satellite orbiting in an elliptical orbit are respectively, 45000 km and 7000 km. Determine the followings: i) Semi-major axis of the elliptical orbit. ii) Orbit eccentricity iii) Distance between the center of earth and the center of elliptical orbit.	10	L3	CO1
	b.	Explain briefly any six orbital parameters required to determine a satellite orbit.	10	L2	CO1
Module – 2					
Q.3	a.	Explain the satellite subsystems.	10	L2	CO2
	b.	Explain the solar energy driven power supply system of a satellite.	10	L2	CO2
OR					
Q.4	a.	Describe the telemetry, telecommand and tracking control monitoring system of a communication satellite.	10	L2	CO2
	b.	Explain with block schematic arrangement of a generalized earth's station.	10	L2	CO2
Module – 3					
Q.5	a.	What is transponder? Explain the various types of transponders.	10	L2	CO3
	b.	List the advantages and disadvantages of satellites with respect to terrestrial networks.	10	L1	CO3
OR					
Q.6	a.	Explain with a neat diagram satellite point-to-point telephonic network.	10	L2	CO3
	b.	Explain with a neat diagram satellite – cable TV.	10	L2	CO3
1 of 2					

Module – 4

Q.7	a.	Explain the mode theory as applied to circular wavelength (wave guides) in optical fibers.	10	L2	CO4
	b.	Describe the operational difference between single-mode and multimode fibers in terms of bandwidth and attenuation.	10	L2	CO4

OR

Q.8	a.	What is modal delay and how does it contribute to modal dispersion in multimode fibers?	10	L2	CO4
	b.	Define material dispersion and explain how it arises in optical fibers.	10	L2	CO4

Module – 5

Q.9	a.	Explain the principle operation of LED's.	10	L2	CO5
	b.	Discuss the characteristics of the optical detectors.	10	L2	CO5

OR

Q.10	a.	Explain the principle operation of WDM standards.	12	L2	CO5
	b.	Explain the isolators and circulators.	8	L2	CO5

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Question Paper Version : A

Fifth Semester B.E./B.Tech Degree Examination, Dec.2024/Jan.2025
Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
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-
1. Which of the following is a Producer?
 a) Animals b) Human beings c) Plants d) Fish
 2. An ecosystem consists of
 a) Biotic species b) Biotic species and physical factors
 c) Only insects d) Only materials
 3. The term Ecology was coined by
 a) E.P. Odium b) A. G. Tansley c) Albert Einstein d) A.G. Haeckel
 4. Which year did the concept of sustainability first appear?
 a) 1992 b) 1978 c) 1980 d) 1987
 5. According to United Nations, the following are the significant issues of social progress.
 a) Instruction b) Public Health c) Living standards d) All of these
 6. The number of SDGS are
 a) 10 b) 15 c) 17 d) 16
 7. Which one of the following is not comes under SDG's?
 a) Zero Hunger b) Political activity c) Education d) Gender quality
 8. Which of the following has the largest population in food chain?
 a) Producers b) 1st degree consumers
 c) 2nd degree consumers d) Decomposers
 9. U.N era of scheduling for sustainable growth is from
 a) 2002 - 11 b) 2003 - 12 c) 2004 - 13 d) 2005 - 14

10. Which of the following is an example of Artificial Ecosystem?
a) Forest b) River c) Aquarium d) Lake
11. Solar Energy is an example of
a) Renewable b) Continuous c) Non - renewable d) None of these
12. Hydrogen energy can be tapped through
a) Heat pumps b) Fuel cells c) Petroleum d) Gassifiers
13. OTEC is an energy technology that converts
a) Energy in large tides of ocean to generate electricity
b) Energy in ocean waves to generate electricity
c) Energy in ocean due to thermal gradient to generate electricity
d) Energy in the fast moving ocean currents to generate electricity.
14. Which of the following source of energy is less eco – friendly?
a) Biogas b) Wind c) Solar d) Nuclear
15. O₃ Represents
a) Oxygen b) Planets c) Ozone d) Green house gases
16. Which of the following is an example of Water Borne Diseases
a) Hepatitis - A b) Hepatitis - B c) Hepatitis - C d) Hepatitis - H
17. Identify the renewable sourced of energy from the following :
a) Coal b) Wind power c) Uranium d) Oil
18. Which of the following is biotic?
a) Forest b) Iron c) Water d) None
19. World Environment Day celebrated on
a) 6th June b) 5th July c) 5th June d) 5th Jan
20. Which of the following is conventional source of Energy?
a) Wind Energy b) Solar Energy c) Bio gas d) None of these
21. Temporary hardness of water is due to
a) Chloride hardness b) Manganese hardness
c) Calcium hardness d) Carbonate hardness
22. The green house gas is
a) N₂O b) CH₄ c) CO₂ d) All of these
23. Which is not a natural source of pollution?
a) Volcanoes b) Forest - Fire c) Coal - fire d) Dust storms
24. Air pollution from automobiles can be controlled by providing
a) Wet collector b) Scrubbers c) Catalytic converter d) All of these
25. Normal sound level of quiet speech is
a) 120 dB b) 90 dB c) 140 dB d) 50 dB

26. What is the permissible range of p^H for drinking water as per the Indian standard?
 a) 6 to 9 b) 6.5 to 7.5 c) 6 to 8.5 d) 6.5 to 8.5
27. Which of the following is a major cause of soil pollution?
 a) Accidents involving the vehicles that are transporting waste material.
 b) Pesticides and chemical fertilizers from agricultural lands
 c) Improper solid waste disposal d) All of these
28. What does E – Waste stand for?
 a) Environment Waste b) Electronic Waste
 c) Equipment Waste d) All of these
29. What is municipal solid waste disposal?
 a) The placement on the land b) The lack of waste
 c) In sanitary land fill site d) All of these
30. What is major cause of air pollution?
 a) Fossil fuel burning b) Industry emission
 c) Wild fire d) All of these
31. Average long – term weather of an area is called
 a) Weather conditions b) Seasonal variations
 c) Average weather d) Climate
32. Climate and global air circulations are mainly affected by the properties of
 a) Water and Air b) Temperature c) Precipitation d) None of these
33. The p^H of acid rain is
 a) less than 5.2 b) 5.2 – 6.2 c) 6.2 – 7.2 d) Above 7.2
34. Acid rain has been increasing day by day due to
 a) Urbanisation b) Industrialisation
 c) Increase in vehicle population d) None of these
35. Which is responsible for Ozone depletion?
 a) Methyl bromide b) CFC's
 c) Hydro chlorofluorocarbons d) All of these
36. The International protocol to protect the Ozone layer's
 a) Kytoto protocol b) Basal protocol c) Montreal protocol d) Viema protocol
37. The Ozone layer is locate upto _____ km above the earth surface.
 a) 1000 b) 50 c) 80 d) 100
38. Acid rain effects on
 a) Materials b) Plants c) Soil d) All of these
39. High – Concentration of fluoride content in drinking water causes.
 a) Dental flurosis b) Bone brittling c) Dental caries d) Plague
40. Desired concentration of fluoride content in drinking water in mg/ℓ is
 a) 0.7 – 1.2 b) 0.5 – 2 c) 2 – 5 d) 1 – 1.75

41. The First United National conference as Human Environment was held at Stickholm in
a) December 1972 b) June 1972 c) June 1974 d) June 1992
42. To achieve the goal of clean environment, important strategies required are
a) Effective laws b) Active participation of the publics
c) Active participation of NGO's d) Both (a) and (b)
43. There are provisions for protection for our environment under
a) Indian Penal Code b) Police Act c) Municipal Act d) All of these
44. The Government of India enacted the Water Act in the year.
a) 1970 b) 1974 c) 1975 d) 1980
45. The Environmental Act of India was enacted in the year.
a) 1986 b) 1992 c) 1984 d) 1974
46. The Environmental Act, 1986 deals with
a) Air b) Water c) Land d) All of these
47. Karnataka State "Pollution Control Board" was established in the year.
a) 1974 b) 1982 c) 1986 d) 1976
48. Which of the following is not a waste treatment method for biomedical waste?
a) Incineration b) Chemical disinfecting
c) Auto claving d) Sieving
49. The Organic material of the solid waste will decompose.
a) By the flow of water b) By the soil particles
c) By the action of micro organisms d) B Oxidation
50. Which of the following element make e – waste hazardous in nature?
a) Lead b) Glass c) Plastic d) Iron

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Question Paper Version : B

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

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46. What is the permissible range of p^H for drinking water as per the Indian standard?
a) 6 to 9 b) 6.5 to 7.5 c) 6 to 8.5 d) 6.5 to 8.5
47. Which of the following is a major cause of soil pollution?
a) Accidents involving the vehicles that are transporting waste material.
b) Pesticides and chemical fertilizers from agricultural lands
c) Improper solid waste disposal d) All of these
48. What does E – Waste stand for?
a) Environment Waste b) Electronic Waste
c) Equipment Waste d) All of these
49. What is municipal solid waste disposal?
a) The placement on the land b) The lack of waste
c) In sanitary land fill site d) All of these
50. What is major cause of air pollution?
a) Fossil fuel burning b) Industry emission
c) Wild fire d) All of these

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Question Paper Version : C

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. Temporary hardness of water is due to
 - a) Chloride hardness
 - b) Manganese hardness
 - c) Calcium hardness
 - d) Carbonate hardness
 2. The green house gas is
 - a) N_2O
 - b) CH_4
 - c) CO_2
 - d) All of these
 3. Which is not a natural source of pollution?
 - a) Volcanoes
 - b) Forest - Fire
 - c) Coal - fire
 - d) Dust storms
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a) December 1972 b) June 1972 c) June 1974 d) June 1992
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c) Active participation of NGO's d) Both (a) and (b)
13. There are provisions for protection for our environment under
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14. The Government of India enacted the Water Act in the year.
a) 1970 b) 1974 c) 1975 d) 1980
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a) 1986 b) 1992 c) 1984 d) 1974
16. The Environmental Act, 1986 deals with
a) Air b) Water c) Land d) All of these
17. Karnataka State "Pollution Control Board" was established in the year.
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18. Which of the following is not a waste treatment method for biomedical waste?
a) Incineration b) Chemical disinfecting
c) Auto claving d) Sieving
19. The Organic material of the solid waste will decompose.
a) By the flow of water b) By the soil particles
c) By the action of micro organisms d) B Oxidation
20. Which of the following element make e – waste hazardous in nature?
a) Lead b) Glass c) Plastic d) Iron
21. Average long – term weather of an area is called
a) Weather conditions b) Seasonal variations
c) Average weather d) Climate
22. Climate and global air circulations are mainly affected by the properties of
a) Water and Air b) Temperature c) Precipitation d) None of these
23. The p^H of acid rain is
a) less than 5.2 b) 5.2 – 6.2 c) 6.2 – 7.2 d) Above 7.2

24. Acid rain has been increasing day by day due to
 - a) Urbanisation
 - b) Industrialisation
 - c) Increase in vehicle population
 - d) None of these
25. Which is responsible for Ozone depletion?
 - a) Methyl bromide
 - b) CFC's
 - c) Hydro chlorofluorocarbons
 - d) All of these
26. The International protocol to protect the Ozone layer's
 - a) Kytoto protocol
 - b) Basal protocol
 - c) Montreal protocol
 - d) Viema protocol
27. The Ozone layer is locate upto _____ km above the earth surface.
 - a) 1000
 - b) 50
 - c) 80
 - d) 100
28. Acid rain effects on
 - a) Materials
 - b) Plants
 - c) Soil
 - d) All of these
29. High – Concentration of fluoride content in drinking water causes.
 - a) Dental flurosis
 - b) Bone brittling
 - c) Dental caries
 - d) Plague
30. Desired concentration of fluoride content in drinking water in mg/ℓ is
 - a) 0.7 – 1.2
 - b) 0.5 – 2
 - c) 2 – 5
 - d) 1 – 1.75
31. Solar Energy is an example of
 - a) Renewable
 - b) Continuous
 - c) Non - renewable
 - d) None of these
32. Hydrogen energy can be tapped through
 - a) Heat pumps
 - b) Fuel cells
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33. OTEC is an energy technology that converts
 - a) Energy in large tides of ocean to generate electricity
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34. Which of the following source of energy is less eco – friendly?
 - a) Biogas
 - b) Wind
 - c) Solar
 - d) Nuclear
35. O₃ Represents
 - a) Oxygen
 - b) Planets
 - c) Ozone
 - d) Green house gases
36. Which of the following is an example of Water Borne Diseases
 - a) Hepatitis - A
 - b) Hepatitis - B
 - c) Hepatitis - C
 - d) Hepatitis - H
37. Identify the renewable soured of energy from the following :
 - a) Coal
 - b) Wind power
 - c) Uranium
 - d) Oil
38. Which of the following is biotic?
 - a) Forest
 - b) Iron
 - c) Water
 - d) None
39. World Environment Day celebrated on
 - a) 6th June
 - b) 5th July
 - c) 5th June
 - d) 5th Jan

40. Which of the following is conventional source of Energy?
a) Wind Energy b) Solar Energy c) Bio gas d) None of these
41. Which of the following is a Producer?
a) Animals b) Human beings c) Plants d) Fish
42. An ecosystem consists of
a) Biotic species b) Biotic species and physical factors
c) Only insects d) Only materials
43. The term Ecology was coined by
a) E.P. Odiyum b) A. G. Tansley c) Albert Einstein d) A.G. Haeckal
44. Which year did the concept of sustainability first appear?
a) 1992 b) 1978 c) 1980 d) 1987
45. According to United Nations, the following are the significant issues of social progress.
a) Instruction b) Public Health c) Living standards d) All of these
46. The number of SDGS are
a) 10 b) 15 c) 17 d) 16
47. Which one of the following is not comes under SDG's?
a) Zero Hunger b) Political activity c) Education d) Gender quality
48. Which of the following has the largest population in food chain?
a) Producers b) 1st degree consumers
c) 2nd degree consumers d) Decomposers
49. U.N era of scheduling for sustainable growth is from
a) 2002 - 11 b) 2003 - 12 c) 2004 - 13 d) 2005 - 14
50. Which of the following is an example of Artificial Ecosystem?
a) Forest b) River c) Aquarium d) Lake

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Question Paper Version : D

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
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BMATEC301/BEC301/BBM301

Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 AV Mathematics III for EC/ BM Engineering

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Statistical table and Mathematics formula handbook are allowed.
3. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1				M	L	C																	
Q.1	a.	Obtain the Fourier series of $f(x) = \frac{\pi - x}{2}$ in $0 < x < 2\pi$. Hence deduce that $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}$.	6	L2	CO1																		
	b.	Find the Fourier series of $f(x) = x $ in $(-\ell, \ell)$. Hence show that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$.	7	L3	CO1																		
	c.	Expand $f(x) = 2x - 1$ as a cosine half range Fourier series in $0 < x < 1$.	7	L2	CO1																		
OR																							
Q.2	a.	Find the Fourier series of $f(x) = \begin{cases} 1 + \frac{2x}{\pi} & \text{in } -\pi < x < 0 \\ 1 - \frac{2x}{\pi} & \text{in } 0 < x < \pi \end{cases}$. Hence deduce that $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$.	6	L2	CO1																		
	b.	Obtain the sine half range series of, $f(x) = \begin{cases} \frac{1}{4} - x & \text{in } 0 < x < \frac{1}{2} \\ x - \frac{3}{4} & \text{in } \frac{1}{2} < x < 1 \end{cases}$	7	L2	CO1																		
	c.	Determine the constant term and the first cosine and sine terms of the Fourier series expansion of y from the following data : <table><tr><td>x°:</td><td>0</td><td>45</td><td>90</td><td>135</td><td>180</td><td>225</td><td>270</td><td>315</td></tr><tr><td>y:</td><td>2</td><td>$\frac{3}{2}$</td><td>1</td><td>$\frac{1}{2}$</td><td>0</td><td>$\frac{1}{2}$</td><td>1</td><td>$\frac{3}{2}$</td></tr></table>	x°:	0	45	90	135	180	225	270	315	y:	2	$\frac{3}{2}$	1	$\frac{1}{2}$	0	$\frac{1}{2}$	1	$\frac{3}{2}$	7	L1	CO1
x°:	0	45	90	135	180	225	270	315															
y:	2	$\frac{3}{2}$	1	$\frac{1}{2}$	0	$\frac{1}{2}$	1	$\frac{3}{2}$															
Module – 2																							
Q.3	a.	Find the Fourier transform of the function, $f(x) = \begin{cases} 1 & \text{for } x \leq a \\ 0 & \text{for } x > a \end{cases}$. Hence evaluate $\int_0^\infty \frac{\sin x}{x} dx$.	6	L2	CO2																		
	b.	Find the Fourier sine and cosine transforms of $f(x) = e^{-\alpha x}$, $\alpha > 0$.	7	L2	CO2																		
	c.	Find the Fourier sine transform of $\frac{e^{-ax}}{x}$, $a > 0$.	7	L3	CO2																		

OR																	
Q.4	a.	If $f(x) = \begin{cases} 1-x^2, & x < 1 \\ 0, & x \geq 1 \end{cases}$, find the Fourier transform of $f(x)$ and hence find the value of, $\int_0^{\infty} \frac{x \cos x - \sin x}{x^3} dx$.	6	L2	CO2												
	b.	Find the Fourier sine transform of $f(x) = e^{- x }$ and hence evaluate $\int_0^{\infty} \frac{x \sin mx}{1+x^2} dx$, $m > 0$.	7	L3	CO2												
	c.	Find the Discrete fast fourier of signal $= (0, 1, 49)^T$	7	L3	CO2												
Module – 3																	
Q.5	a.	Find the z-transform of, (i) $\cosh n\theta$ (ii) $\sinh n\theta$	6	L1	CO3												
	b.	If $V(z) = \frac{2z^2 + 3z + 12}{(z-1)^4}$, evaluate u_0, u_1 and u_2	7	L2	CO3												
	c.	Find the inverse z-transform of, $\frac{z}{(z-1)(z-2)}$.	7	L2	CO3												
OR																	
Q.6	a.	Solve by using z-transforms, $y_{n+2} + 2y_{n+1} + y_n = n$ with $y_0 = 0 = y_1$	6	L3	CO3												
	b.	Find $z^{-1} \left[\frac{5z}{(3z-1)(2-z)} \right]$.	7	L2	CO3												
	c.	Solve by using z-transforms $u_{n+2} - 5u_{n+1} + 6u_n = 2^n$ with $u_0 = 0 = u_1$.	7	L3	CO3												
Module – 4																	
Q.7	a.	Solve $\frac{d^3 y}{dx^3} + 6\frac{d^2 y}{dx^2} + 11\frac{dy}{dx} + 6y = 0$.	6	L1	CO4												
	b.	Solve $(D^2 + 1)y = x^2 + 4x - 6$.	7	L2	CO4												
	c.	Using the method of variation of Parameters of $\frac{d^2 y}{dx^2} - 6\frac{dy}{dx} + 9y = e^{3x}$	7	L3	CO4												
OR																	
Q.8	a.	Solve $6\frac{d^2 y}{dx^2} + 17\frac{dy}{dx} + 12y = e^{-x}$.	6	L2	CO4												
	b.	Solve the Cauchy's differential equation, $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + 8y = 65 \cos(\log x)$.	7	L2	CO4												
	c.	The charge q in a series circuit containing an Inductance L , Capacitance C , emf E satisfy the differential equation, $L \frac{d^2 q}{dt^2} + \frac{q}{C} = E$. Express q in terms of t .	7	L3	CO4												
Module – 5																	
Q.9	a.	Fit a second degree parabola $y = a + bx + cx^2$ into least square sense for the data and estimate y at $x = 6$. <table border="1" data-bbox="331 1809 667 1883"> <tr> <td>x:</td> <td>1</td> <td>2</td> <td>3</td> <td>4</td> <td>5</td> </tr> <tr> <td>y:</td> <td>10</td> <td>12</td> <td>13</td> <td>16</td> <td>19</td> </tr> </table>	x:	1	2	3	4	5	y:	10	12	13	16	19	6	L1	CO5
x:	1	2	3	4	5												
y:	10	12	13	16	19												

	b.	Find a correlation coefficient for the two variables x and y. <table><tr><td>x:</td><td>92</td><td>89</td><td>87</td><td>86</td><td>83</td><td>77</td><td>71</td><td>63</td><td>53</td><td>50</td></tr><tr><td>y:</td><td>86</td><td>83</td><td>91</td><td>77</td><td>68</td><td>85</td><td>52</td><td>82</td><td>37</td><td>57</td></tr></table>	x:	92	89	87	86	83	77	71	63	53	50	y:	86	83	91	77	68	85	52	82	37	57	7	L2	CO5
x:	92	89	87	86	83	77	71	63	53	50																	
y:	86	83	91	77	68	85	52	82	37	57																	
	c.	Ten students got the following percentage of marks in two subjects x and y. Compute the rank correlation coefficient. <table><tr><td>x:</td><td>78</td><td>36</td><td>98</td><td>25</td><td>75</td><td>82</td><td>90</td><td>62</td><td>65</td><td>39</td></tr><tr><td>y:</td><td>84</td><td>51</td><td>91</td><td>60</td><td>68</td><td>62</td><td>86</td><td>58</td><td>53</td><td>47</td></tr></table>	x:	78	36	98	25	75	82	90	62	65	39	y:	84	51	91	60	68	62	86	58	53	47	7	L2	CO5
x:	78	36	98	25	75	82	90	62	65	39																	
y:	84	51	91	60	68	62	86	58	53	47																	
OR																											
Q.10	a.	If θ is the angle between the lines of regression show that $\tan \theta = \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2} \left(\frac{1 - r^2}{r} \right).$	6	L2	CO5																						
	b.	Obtain the lines of regression and hence find the coefficient of correlation for the data, <table><tr><td>x:</td><td>1</td><td>3</td><td>4</td><td>2</td><td>5</td><td>8</td><td>9</td><td>10</td><td>13</td><td>15</td></tr><tr><td>y:</td><td>8</td><td>6</td><td>10</td><td>8</td><td>12</td><td>16</td><td>16</td><td>10</td><td>32</td><td>32</td></tr></table>	x:	1	3	4	2	5	8	9	10	13	15	y:	8	6	10	8	12	16	16	10	32	32	7	L2	CO5
x:	1	3	4	2	5	8	9	10	13	15																	
y:	8	6	10	8	12	16	16	10	32	32																	
	c.	If $8x - 10y + 66 = 0$ and $40x - 18y = 214$ are the two regression lines. Find $\bar{x}$, $\bar{y}$ and r. Find σ_y if $\sigma_x = 3$.	7	L2	CO5																						

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Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Research Methodology and IPR

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Identify the meaning of Research and brief out the objective and motivation in engineering research.	10	L1	CO1
	b.	Explain brief about research cycle and verify with the research flow diagram.	10	L1	CO1
OR					
Q.2	a.	Identify the types of engineering research and briefly explain them.	10	L1	CO1
	b.	Explain about the different types of research misconduct.	10	L1	CO1
Module – 2					
Q.3	a.	Explain about the importance of literature review and technical reading.	10	L2	CO2
	b.	Mention the various benefits of bibliographic databases.	10	L1	CO2
OR					
Q.4	a.	Identify the impact of technical reaction and brief about it.	10	L1	CO2
	b.	Enumerate the impact of title and keywords on citation with example.	10	L2	CO2
Module – 3					
Q.5	a.	Define Intellectual properties and explain about its types.	10	L1	CO3
	b.	Explain about the key aspect of patent law.	10	L2	CO3
OR					
Q.6	a.	Explain about the assessment of novelty.	10	L1	CO3
	b.	Brief about the patent procedure in India.	10	L1	CO4
Module – 4					
Q.7	a.	Mention and brief about the justification for copyright law.	10	L2	CO4
	b.	Explain about the basic concepts of under lying copyright law.	10	L1	CO4
OR					
Q.8	a.	Brief about the various representations of sound recordings.	10	L2	CO5
	b.	Explain about TRIPS agreement in detail.	10	L1	CO5

Module – 5					
Q.9	a.	Explain about the justification of protection designs.	10	L2	CO5
	b.	Brief about the excluded subjected matter in the context of design protection.	10	L1	CO5
OR					
Q.10	a.	What are the rights of the owner of designs? Explain.	10	L1	CO5
	b.	Brief about the Assignment of Design Rights.	10	L1	CO5

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21EC51

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Digital Communication

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With necessary diagram, explain the generation and detection or reception of BPSK signal. (08 Marks)
- b. Derive the expression for error probability of BFSK. (08 Marks)
- c. Define bandwidth efficiency. Tabulate and comment on the bandwidth efficiency of m-ary PSK. (04 Marks)

OR

- 2 a. Sketch QPSK waveform for the binary data 01101000. (08 Marks)
- b. A binary FSK system transmits binary data at a rate of 2 Mbps over AWGN channel. The noise power spectral density $\left(\frac{N_0}{2}\right) = 10^{-20}$ W/Hz. Determine the probability of error for coherent detection of FSK scheme. Assume the amplitude of the received signal as 1 μ v. Consider $\text{erf}(2.5) = 0.99959$ or $\text{erfc}(\sqrt{625}) = 0.00041$. (06 Marks)
- c. With a neat block diagram, explain the generation of DPSK signal. (06 Marks)

Module-2

- 3 a. For the signals $s_1(t)$, $s_2(t)$, $s_3(t)$, shown in the given Fig.Q3(a), find the set of orthonormal basis function using GSOP. (10 Marks)

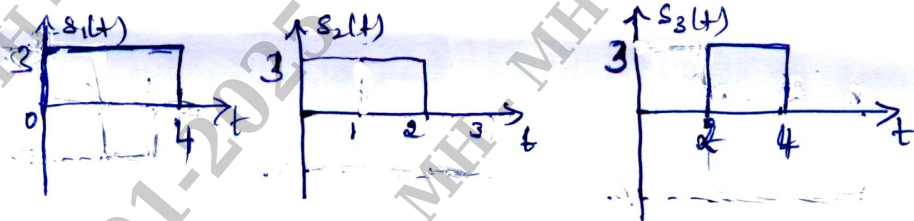


Fig.Q3(a)

- b. Explain the matched filter receiver with the relevant mathematical expressions. (06 Marks)
- c. Explain how to convert continuous AWGN channel into a vector channel. (04 Marks)

OR

- 4 a. Explain the design of band limited signals with controlled ISI, describe the Time domain and frequency domain characteristics of a duo-binary signal. (08 Marks)
- b. The binary sequence 111010010001101 is the input to the precoder whose output is used to modulate a duo binary transmitter filter. Obtain the precoded sequence, transmitted amplitude levels, the received signal level and the decoded sequence. (08 Marks)
- c. State Nyquist criteria. (04 Marks)

Module-3

- 5 a. Explain the generation of direct sequence spread spectrum with relevant waveform and spectrum. (08 Marks)
- b. Explain any three applications of DSSS. (06 Marks)
- c. List and explain the properties of PN sequence. (06 Marks)

OR

- 6 a. With a neat block diagram, explain the frequency hopped spread spectrum. (08 Marks)
- b. Draw a 3-stage LFSR, with first and 3rd stage connected to a modulo 2 adder and the output sequence is given by the 3rd stage. Consider 110 as the initial state. (08 Marks)
- c. The spread spectrum communication system has the following parameters, $T_b = 1.024$ msec, PN chip duration of 1 μ sec. The average probability of error of system is not to exceed 10^{-5} . Calculate length of shift register, processing gain and Jamming margin. (04 Marks)

Module-4

- 7 a. A code is composed of dots and dashes. Assuming that a dash is 3 times as long as a dot and has 1/3 the probability of occurrence, calculate:
- The information in a dot and a dash
 - Entropy of dot dash code
 - Average rate of information, if a dot lasts for 10 msec and this time is allowed between symbols. (08 Marks)
- b. Given the message x_1, x_2, x_3, x_4, x_5 and x_6 with respective probabilities 0.4, 0.2, 0.2, 0.1, 0.07 and 0.03. Construct a binary code by applying Shannon's fano encoding procedure and determine the code efficiency and redundancy. (08 Marks)
- c. Define the following with respect to information theory:
- Self information
 - Rate of information (04 Marks)

OR

- 8 a. Apply Shannon's encoding binary algorithm to the following set of messages and obtain code efficiency and redundancy.

m_1	m_2	m_3	m_4	m_5
$\frac{1}{8}$	$\frac{1}{16}$	$\frac{3}{16}$	$\frac{1}{4}$	$\frac{3}{8}$

- b. Given the messages s_1, s_2, s_3 and s_4 with respective probabilities of 0.4, 0.3, 0.2 and 0.1. Construct a binary code by applying Huffman encoding procedure determine code efficiency and redundancy of the code. (08 Marks)
- c. List and explain the error control codes. (04 Marks)

Module-5

- 9 a. Consider a (6, 3) linear code where generator matrix is :

$$h = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

- Find all code vector.
- Find all the hamming weight and distances.
- Find min weight parity check matrix.
- Draw the encoder circuit for the above codes. (10 Marks)

- b. For a systematic (7, 4) linear block code, the parity matrix 'P' is given by

$$P = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

- Find all possible code vector.
- Draw the corresponding encoding circuit.
- A single error has occurred in each of these received vector, detect and correct those errors.

$$R_A = [0111110]$$

$$R_B = [1011100]$$

$$R_C = [1010000]$$

(10 Marks)

OR

- 10 a. For the convolution encoder shown in Fig.Q10(a), the information sequence is $d = 10011$. Find the o/p sequence using the following 2 approaches.

- Time domain approach
- Frequency domain approach/transform domain approach

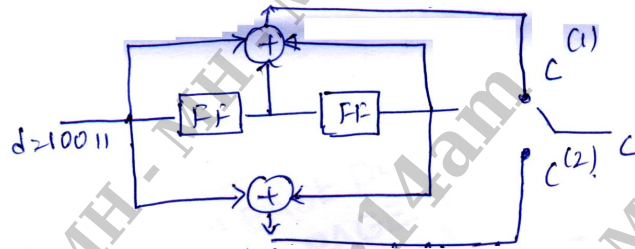


Fig.Q10(a) (2, 1, 2) convolutional encoder

(10 Marks)

- b. A rate 1/3 convolutional encoder has generating vectors $g_1 = 111$, $g_2 = 101$.
- Sketch the encoder configuration, write the transition table.
 - Draw the code tree and state diagram.
 - If input message sequence is 10111, determine the output sequence of the encoder using transform domain approach.

(10 Marks)

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21EC52

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Computer Organization and ARM Microcontrollers

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the pipelining and superscalar operation. (05 Marks)
- b. Briefly explain the different key parameters that affects the processor performance. (05 Marks)
- c. With a neat diagram, explain basic operational concept of computer. (10 Marks)

OR

- 2 a. Mention the difference between Big-endian and Little-endian assignments. (05 Marks)
- b. Explain the different types of addressing modes. (10 Marks)
- c. With diagram explain the Interrupt Hardware. (05 Marks)

Module-2

- 3 a. With diagram explain the internal organization of a 2M×8 dynamic memory chip. (10 Marks)
- b. Explain the operations of synchronous DRAM. (10 Marks)

OR

- 4 a. Explain the different types of memories. (10 Marks)
- b. With neat diagram explain the multiple bus organization. (10 Marks)

Module-3

- 5 a. Explain the architecture of ARM core dataflow model. (10 Marks)
- b. With neat diagram explain the ARM based embedded system. (10 Marks)

OR

- 6 a. Explain the ARM condition flag register. (05 Marks)
- b. What is pipeline? Explain the 3-stage ARM pipeline. (05 Marks)
- c. Explain the architecture of ARM processor. (10 Marks)

Module-4

- 7 a. With example explain the following instructions:
i) ADC ii) EOR iii) SWI iv) UMULL v) SBC (10 Marks)
- b. With neat diagram explain the ARM stack operation. (10 Marks)

OR

- 8 a. Explain the ARM Registers and also explain each. (10 Marks)
- b. Explain the following ARMVSE extension instructions:
i) CLZ ii) QADD iii) QSUB iv) SMLAxy v) QDADD (10 Marks)

Module-5

- 9 a. Explain the following THUMB instructions:
i) TST ii) ROR iii) BX iv) BKPT v) ASR (10 Marks)
- b. Discuss the ARM support basic C data types. (10 Marks)

OR

- 10 a. Explain the ARM function calls and loop operations. (10 Marks)
- b. With example explain the single register and multiple register load store ARM instructions. (10 Marks)

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Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and /or equations written eg, 42+8 = 50, will be treated as malpractice.

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21EC53

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Computer Communication Networks

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Describe the layers of the TCP/IP protocol suite and explain the service provided by each layer with the help of a neat diagram. (10 Marks)
- b. With neat diagrams, explain the working of ARP. Describe the structure and functions of various fields in the ARP packet. (10 Marks)

OR

- 2 a. With neat diagrams, explain the salient features, advantages and disadvantages of various physical topologies used in computer networks. (12 Marks)
- b. Explain with neat diagrams, the concepts of encapsulation and decapsulation at various layers of the TCP/IP protocol suite, during communication between two hosts. (08 Marks)

Module-2

- 3 a. With clear examples, illustrate the following concepts and explain:
 - (i) Byte stuffing and unstuffing
 - (ii) Bit stuffing and unstuffing(10 Marks)
- b. An ALOHA network transmits 200 bit frames on a shared channel of 200 kbps. Calculate the throughput of the system if it produces 1000 frames per second in case of:
 - (i) Pure ALOHA
 - (ii) SLOTTED ALOHA(06 Marks)
- c. Explain the Ethernet Frame Format with a neat diagram. (04 Marks)

OR

- 4 a. With relevant flow diagrams, explain 1-persistent, non-persistent and p-persistent methods in CSMA. (10 Marks)
- b. Explain the working of CSMA/CA protocol with a neat flow diagram. Describe how CSMA/CA overcomes the problems of collision during handshaking and hidden stations. (10 Marks)

Module-3

- 5 a. Explain virtual circuit approach used in packet switching. With neat diagrams and an example, illustrate the 3 stages of virtual circuit approach. (10 Marks)
- b. With a neat diagram, explain the IPv4 datagram. (10 Marks)

OR

- 6 a. Explain distance vector routing algorithm, using Bellman-Ford equations. Illustrate the same with an example. (10 Marks)
- b. Describe how the IPV4 address space is occupied in classful addressing. (04 Marks)
- c. An organization is granted a block of addresses with the beginning address 14.24.74.0/24. The organization needs to have 3 subblocks of addresses to use in its 3 subnets: one subblock of 10, one subblock of 60, and one subblock of 120 addresses. Design the subblocks. (06 Marks)

Module-4

- 7 a. Describe the connectionless and connection-oriented services provide by the transport layer in TCP/IP. (10 Marks)
- b. With a neat diagram, explain the TCP segment format, including the pseudoheader. (10 Marks)

OR

- 8 a. Explain the selective repeat protocol with neat diagrams and illustrate with an example. (10 Marks)
- b. The content of a UDP header format is given as CB84000D001C001C. Determine the following:
- (i) The source port number
 - (ii) The destination port number
 - (iii) Total length of the user datagram
 - (iv) Length of the data
 - (v) Is the packet directed from client to server or vice-verse? (05 Marks)
- c. Explain Checksum calculation in UDP using pseudoheader. (05 Marks)

Module-5

- 9 a. Explain the architecture of Electronic Mail, with a neat diagram. (10 Marks)
- b. Explain persistent and non-persistent connections in HTTP, with example. (10 Marks)

OR

- 10 a. With a neat taxonomy, describe various security attacks in communication networks. (08 Marks)
- b. With neat diagrams, explain the following with respect to DNS:
- (i) Name space
 - (ii) DNS in the internet
 - (iii) Name Address Resolution (12 Marks)

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21EC54

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Electromagnetic Waves

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. If two position vectors $\vec{A} = -2\vec{a}_x - 5\vec{a}_y - 4\vec{a}_z$ and $\vec{B} = 2\vec{a}_x + 3\vec{a}_y + 5\vec{a}_z$ then find,
i) $\vec{AB}$ ii) $\vec{a}_A, \vec{a}_B$ iii) $\vec{a}_{AB}$ iv) Unit vector from C to A where C is (3, 5, 8) (06 Marks)
- b. Ten identical charges each of 500 μC are spaced equally around a circle of radius 2 m. Find the force on a charge of $-20 \mu\text{C}$ located on the axis, 2m from the plane of the circle. (07 Marks)
- c. Define Electric Field Intensity. Derive expression for electric field intensity due to 'n' number of charges. (06 Marks)

OR

- 2 a. Given the two points A(2, 3, -1) and B(4, 25, 120). Find spherical coordinates of A and Cartesian coordinates of B. (06 Marks)
- b. Derive an expression for electric field intensity due to infinite line charge. (07 Marks)
- c. Find electric field $\vec{E}$ at origin, if the following charge distribution are present in free space :
i) Point charge of 21nC at P(2, 0, 6)
ii) Uniform line charge of infinite length with charge density $\rho_L = 3 \text{ nC/m}$ at $x = 2, y = 3$.
iii) Uniform surface charge of density 0.2 nC/m^2 at $x = 2$. (07 Marks)

Module-2

- 3 a. A charge is uniformly distributed over a spherical surface of radius 'a'. Determine electric field intensity at all the places, use Gauss law. (07 Marks)
- b. Evaluate both sides of divergence theorem for the field $\vec{D} = 2xy\vec{a}_x + x^2\vec{a}_y \text{ C/m}^2$, for a rectangular parallel piped formed by the planes $x = 0$ and $x = 1$; $y = 0$ and $y = 2$; $z = 0$ and $z = 3$. (08 Marks)
- c. Show that electric field intensity is negative potential gradient. (05 Marks)

OR

- 4 a. The flux density $\vec{D} = \frac{r}{3} \vec{a}_r \text{ nC/m}^2$ in free space :
i) Find $\vec{E}$ at $r = 0.2 \text{ m}$
ii) Find the total electric flux leaving the sphere of $r = 0.2 \text{ m}$.
iii) Find the total charge within the sphere of $r = 0.3 \text{ m}$ (07 Marks)
- b. State and prove Gauss divergence theorem. (07 Marks)
- c. Derive an expression for continuously equation. (06 Marks)

Module-3

- 5 a. Determine whether or not the following potential fields satisfy the Laplace's equation:
 i) $V = x^2 - y^2 + z^2$ ii) $V = r \cos\phi + z$ iii) $V = r \cos\theta + \phi$ (06 Marks)
- b. Evaluate both sides of Stoke's theorem for the field $\vec{H} = 6xy\vec{a}_x - 3y^2\vec{a}_y$ A/m, for rectangular path around the region, $2 \leq x \leq 5$, $-1 \leq y \leq 1$, $z = 0$. (08 Marks)
- c. Explain the concept of magnetic potential. (06 Marks)

OR

- 6 a. State and prove uniqueness theorem. (06 Marks)
- b. Two plates of parallel plate capacitor are separated by distance 'd' and maintained at potential zero and V_0 respectively. Determine,
 i) Potential at any position between the plates
 ii) Surface charge density on the plates
 iii) Capacitance between the plates. (08 Marks)
- c. Find the magnetic flux density at the centre 'O' of a square of sides equal to 5 m and carrying 10 A of current. (06 Marks)

Module-4

- 7 a. Derive an expression for force on a differential current element and find force experienced by conductor of 6 m long, lies along z-direction with a current of 2A in $\vec{a}_z$ direction, if $\vec{B} = 0.08\vec{a}_x$ T. (07 Marks)
- b. Explain magnetization and permeability. (07 Marks)
- c. Derive boundary conditions at the interface of two magnetic materials. (06 Marks)

OR

- 8 a. A point charge of $Q = -1.2$ C has velocity $\vec{v} = (5\vec{a}_x + 2\vec{a}_y - 3\vec{a}_z)$ m/s. Find the magnitude of force exerted on the charge if,
 i) $\vec{E} = -18\vec{a}_x + 5\vec{a}_y - 10\vec{a}_z$ V/m
 ii) $\vec{B} = -4\vec{a}_x + 4\vec{a}_y + 3\vec{a}_z$ T
 iii) Both are present simultaneously. (07 Marks)
- b. Find the magnetization in a magnetic material where:
 i) $\mu = 1.8 \times 10^5$ H/m and $M = 120$ A/m
 ii) $\mu = 22$, there are 8.3×10^{28} atoms/m³ each atom has a dipole moment of 4.5×10^{-27} A/m² and
 iii) $B = 300$ μ T and $X_m = 15$ (07 Marks)
- c. State and explain Faraday's law of electromagnetic induction. Hence obtain Maxwell's equation in point form and integral form. (06 Marks)

Module-5

- 9 a. Derive Maxwell's equations for time varying fields, represent them in point form and integral form. (08 Marks)
- b. Obtain relationship between $\vec{E}$ and $\vec{H}$ in free space. (06 Marks)

- c. In free space $\vec{E} = 50 \cos(\omega t - \beta z) \vec{a}_x$ V/m. Find the average power crossing a circular area of radius 2.5m in the plane $z = 0$. (06 Marks)

OR

- 10 a. Given $\vec{E} = E_m \sin(\omega t - \beta z) \vec{a}_y$ in free space, find $\vec{D}$, $\vec{B}$ and $\vec{H}$. Sketch $\vec{E}$ and $\vec{H}$ at $t = 0$. (08 Marks)
- b. Explain wave propagation in good conductor with relevant equations. (08 Marks)
- c. Wet marshy soil is characterized by $\sigma = 10^{-2}$ S/m, $\epsilon_r = 15$ and $\mu_r = 1$. At 1 MHz whether soil be considered as conductor or dielectric. (04 Marks)

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21EC71

Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Advanced VLSI

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain ASIC design flow with neat flow chart. (10 Marks)
- b. Explain Booth Encoding multiplier with an example. (10 Marks)

OR

- 2 a. Describe different cell compilers and I/O cells. (10 Marks)
- b. With a neat diagram, explain the operation of conditional sum adder. Mention its advantages and disadvantages. (10 Marks)

Module-2

- 3 a. Explain the concept of measurement of delay in Floorplanning. (10 Marks)
- b. Explain the following : (10 Marks)
 - i) Power planning
 - ii) Clock planning

OR

- 4 a. Write an algorithm for iterative placement improvement method and explain briefly. (10 Marks)
- c. Explain the goals and objectives of global routing in detail. (10 Marks)

Module-3

- 5 a. Explain factors in randomizing the stimulus to a design. (10 Marks)
- b. Draw the diagram of layered test bench of system verilog and describe the function of each layer. (10 Marks)

OR

- 6 a. Describe the various array methods with an example. (08 Marks)
- b. Describe type def and enumerated data types with example. (06 Marks)
- c. Explain constant and strings in system verilog with example. (06 Marks)

Module-4

- 7 a. Explain Tasks, functions and void function in system verilog. (06 Marks)
- b. How time values are specified in system verilog, describe with example. (06 Marks)
- c. Explain automatic storage and variable initialization with system verilog program example. (08 Marks)

OR

- 8 a. Describe the communication between the test bench and DUT with suitable diagram and system verilog program. (10 Marks)
- b. Explain different types of system verilog assertions with example. (10 Marks)

Module-5

- 9 a. Explain common randomization problems in brief. (10 Marks)
b. Write any 4 Random Number functions with an example. (04 Marks)
c. Explain Pseudorandom Number Generators (PRNG) in system verilog. (06 Marks)

OR

- 10 a. What is coverage? Explain coverage types in system verilog. (10 Marks)
b. Describe various coverage options with an example. (10 Marks)

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21EC72

Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Optical and Wireless Communication

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Briefly explain with a neat figure the propagation mechanism of meridional rays in an ideal step index optical waveguide. (08 Marks)
- b. Define the term attenuation in optical fibers. Explain the different attenuation mechanisms in optical fibers. (12 Marks)

OR

- 2 a. Define Dispersion. Briefly explain intermodal and intramodal dispersion effects in optical waveguide. (10 Marks)
- b. With neat figures, discuss the structure of single mode and multimode step-index and graded index optical fibers. (06 Marks)
- c. A multimode fiber has a core refractive index of 1.480 and a core cladding index difference of 2.0 percent. Find the numerical aperture and critical angle at the core cladding interface. (04 Marks)

Module-2

- 3 a. What are the characteristic requirements of an optical source? With the help of neat diagram, explain the constructional features and emission pattern of surface emitting LED. (10 Marks)
- b. Define optical isolator. With a neat figure, explain the design and operation of a polarization independent isolator. (06 Marks)
- c. A given silicon avalanche photodiode has a quantum efficiency of 65 percent at a wavelength of 900 nm. If 0.5 μW of optical power produces a multiplied photocurrent of 10 μA . What is the multiplication M? (04 Marks)

OR

- 4 a. Discuss the operation of pin photodiode with a neat circuit and energy band diagram. (10 Marks)
- b. What is Diffraction gratings? Discuss briefly Diffraction grating techniques. (10 Marks)

Module-3

- 5 a. Explain briefly the different propagation mechanisms that influence the signal propagation in a mobile communication environment. (10 Marks)
- b. A cellular communication service area is covered with 12 clusters having 7 cells in each cluster and 16 channels assigned in each cell. Find the number of channels per cluster and the system capacity. (03 Marks)
- c. Explain how the concept of frequency reuse increases the spectrum efficiency that in turn increases the cellular communication system capacity. (07 Marks)

OR

- 6 a. Briefly discuss the generations of wireless communication network technology. (08 Marks)
b. Discuss the effects of co-channel interference in wireless communication in reducing the system capacity. (05 Marks)
c. Discuss the concept of multipath fading in mobile communication system. (07 Marks)

Module-4

- 7 a. With a neat block diagram, explain the operation of basic TDMA link. (10 Marks)
b. Explain the basic cellular system with necessary block diagram. (10 Marks)

OR

- 8 a. Discuss with a neat figure the call processing in a cellular system for mobile-originated calls. (12 Marks)
b. List the advantages of CDMA over TDMA and FDMA. (08 Marks)

Module-5

- 9 a. What is Hand off in GSM networks? Explain briefly the different handoff procedure in GSM. (10 Marks)
b. Explain the functions of data bases HLR and VLR at MSC in GSM network architecture and also explain how it is helpful in location updation in GSM networks. (10 Marks)

OR

- 10 a. Briefly explain the three major subsystems in GSM network architecture with a neat block diagram. (10 Marks)
b. Explain briefly the following identifiers in GSM system:
(i) SIM
(ii) Mobile system ISDN with frame format
(iii) Location Area Identify (10 Marks)

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21EC732

Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Digital Image Processing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With neat diagram explain Single image sensor, how it can be used in Sensor Strip and Sensor Array. (08 Marks)
- b. Explain basic concept of Sampling and Quantization with reference to Digital Image. (07 Marks)
- c. Calculate the photon energy for visible light for given wavelength range 400 nm to 750 nm. [Plank's constant, $h = 6.63 \times 10^{-34}$ Js, $C = 3 \times 10^8$ m/s] (05 Marks)

OR

- 2 a. Explain the Brightness Adaption and Discrimination. (07 Marks)
- b. Explain the Neighbour pixel basic relationship in Digital Images with adjacency connectivity, Regions and Boundaries. (08 Marks)
- c. Given two pixels P and Q with coordinate positions (-2, -2) and (3, 4) respectively, calculate the distance measure D_e , D_4 , D_8 . (05 Marks)

Module-2

- 3 a. Define 2-D orthogonal and unitary transform. (06 Marks)
- b. For given orthogonal matrix A and an image u obtain unitary transform.
Given $A = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ $u = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ (08 Marks)
- c. Define the properties of unitary transform. (06 Marks)

OR

- 4 a. Define 2-D DFT and its properties. (06 Marks)
- b. Define cosine transform and its properties. (06 Marks)
- c. Calculate Haar transform for $N = 4$

$$\text{Given Haar function } H_a(z) = \frac{1}{\sqrt{N}} \begin{cases} +2^{p/2} & , \quad \frac{q-1}{2^p} \leq z < \frac{q-0.5}{2^p} \\ -2^{p/2} & , \quad \frac{q-0.5}{2^p} \leq z < \frac{q}{2^p} \\ 0 & , \quad \text{else} \end{cases}$$

$$n = \log_2 N$$

$$p = 0 \text{ to } n-1$$

$$q \text{ range between } 1 \leq q \leq 2^p$$

$$k = 2^p + q - 1$$

$$z = 0, 1/4, 2/4, 3/4$$

(08 Marks)

Module-3

- 5 a. With necessary graph and equation explain
 i) Image Negative
 ii) Power law transformation
 iii) Intensity level slicing (06 Marks)
- b. Compute Histogram equalization for given data:

Table 5(b)

r_k	0	1	2	3	4	5	6	7
n_k	790	1023	850	656	329	245	122	81

for 3 bit image ($L = 8$) of size 64×64 pixels ($MN = 4096$) with intensity distribution shown in Table 5(b). Intensity level are integer in range $[0, L-1] = [0, 7]$ (08 Marks)

- c. With an example for 2-bit image of size 5×5 define the sample mean, sample variance with equation. (06 Marks)

OR

- 6 a. Explain with example fundamentals of Spatial Filtering for spatial correlation and convolution for 1-D and 2-D filter. (08 Marks)
- b. Using 1st order derivative Image Sharpening (the Gradient) define:
 i) Robert's cross gradient operation
 ii) Sobel's operators (for 3×3 region) (06 Marks)
- c. Define smoothing spatial filters with brief note:
 i) Linear Filters
 ii) Order Statistic Filter (06 Marks)

Module-4

- 7 a. With neat block diagram of Homomorphic system, derive Homomorphic filtering approach for Image Enhancement. (08 Marks)
- b. Define sharpening of images in frequency domain using
 i) Ideal High Pass Filter
 ii) Butterworth High Pass Filter
 iii) Gaussian High Pass Filter (06 Marks)
- c. Give Frequency domain filtering necessary steps followed. (06 Marks)

OR

- 8 a. Define pseudo color image processing with intensity slicing and intensity to color transformation. (06 Marks)
- b. Based on Hardware oriented models classify different color model given color conversion for RGB to HIS and vice versa with relevant equation. (08 Marks)
- c. With color fundamentals for primary and secondary colors. (06 Marks)

Module-5

- 9 a. Write brief note on restoration in presence of only noise using
 i) Mean filter ii) Order statistic filter iii) Adaptive filter (08 Marks)
- b. Discuss some of the important noise probability density functions. (06 Marks)
- c. With help of block diagram give details of Degradation / Restoration process. (06 Marks)

OR

- 10 a. In digital images discuss about Inverse Filtering. (06 Marks)
- b. Explain minimum mean square error (Wiener Filter) in Digital Image Processing. (08 Marks)
- c. Discuss periodic noise reduction by frequency domain filtering. (06 Marks)

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Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

DSP Algorithms and Architectures

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Mention and describe the features to be considered in designing and implementing a DSP system. (06 Marks)
- b. Determine the interpolated sequences $y(n)$ with input sequence $x(n) = \{0, 3, 6, 9, 12\}$ using an interpolation filter $b_k = \left[\frac{1}{3}, \frac{2}{3}, 1, \frac{2}{3}, \frac{1}{3} \right]$ and interpolation factor $L = 3$. (06 Marks)
- c. Outline a DSP system with the help of block diagram and its typical signals. (08 Marks)

OR

- 2 a. Discuss the major features of programmable Digital signal processes. (08 Marks)
- b. Evaluate Decimation and interpolation process with relevant equations. (08 Marks)
- c. Determine complex multiplies for Direct DFT and FFT based DFT computations for $N = 4$. (04 Marks)

Module-2

- 3 a. Discuss MAC limit with Guard bits and saturation logic. (10 Marks)
- b. Explain Braun multiplier and Barrel shifter. (10 Marks)

OR

- 4 a. Illustrate special addressing modes of DSP system. (10 Marks)
- b. Illustrate single MAC pipelined implementation of an 8-tap filter with relevant diagrams. (10 Marks)

Module-3

- 5 a. Relate Direct and Indirect addressing modes with suitable block diagram for TMS320C54XX processors. (08 Marks)
- b. What are PMST and TCR registers? Explain with their bit format. (06 Marks)
- c. Discuss the following instructions
i) MAS ii) MACD iii) RPTB. (06 Marks)

OR

- 6 a. Write a program to compute the sum of product term given by the equation
 $y(n) = h_0x(n) + h_1x(n-1) + h_2x(n-2)$
use direct addressing mode to access the data memory. (10 Marks)
- b. Describe pipelining operation in TMS320C54XX processor. Give pipeline operation of given sequence
LD *AR3+, A
ADD#1000h, A
STL A, *AR3 +
.
.
.
(10 Marks)

Module-4

- 7 a. Write a program to implement the FIR filter for TMS320C54XX processors. (10 Marks)
b. Why polyphase subfilters are required in interpolation filters? Explain Digital interpolation implementation using five polyphase subfilters with interpolation factor 5. (10 Marks)

OR

- 8 a. Describe the following with FFT computation
i) Overflow and scaling
ii) Bit – relevant Index Generation (10 Marks)
b. Design and implement a general DIT – FFT butterfly in place computation. Give its structure. (10 Marks)

Module-5

- 9 a. Interface the TMS320C54XX to a 10 bit ADC (TLC1550) and an 8-bit DAC (TLC7524). The sampled signal read from the ADC is to be written to the DAC after adjusting its size. The start of the conversion is to be initiated by the TOUT signal of the timer. (12 Marks)
b. Outline Register subaddressing technique for configuration DMA operation. (08 Marks)

OR

- 10 a. Describe the block diagram for the PCM3002 codec. (10 Marks)
b. Discuss with suitable block diagram JPEG encoding and JPEG decoding process. (10 Marks)

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Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 IoT and Wireless Sensor Networks

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define IoT. Explain the characteristics of IoT. (06 Marks)
- b. Explain the different communication models in logical design of IoT. (08 Marks)
- c. Explain the IoT level-1, IoT level-2 IoT level-3 with relevant block diagrams. (06 Marks)

OR

- 2 a. With respect to physical design of the IoT, explain the following :
i) Things in IoT
ii) IoT protocols. (08 Marks)
- b. Explain any three IoT enabling technologies. (06 Marks)
- c. Explain the IoT level-5, and IoT level-6 with neat block diagram. (06 Marks)

Module-2

- 3 a. Explain the following applications of IoT with respect to home automation :
i) Smart lighting
ii) Smart appliances
iii) Intrusion detection
iv) Smoke/gas detectors. (10 Marks)
- b. Analyze the applications of IoT in environment domain. (10 Marks)

OR

- 4 a. Explain the different applications of IoT for cities. (10 Marks)
- b. Analyze the applications of IoT in the following domains :
i) Agriculture
ii) Industry. (10 Marks)

Module-3

- 5 a. Explain the applications of the sensor networks. (06 Marks)
- b. Mention the characteristic features of sensor networks. With a neat block diagram explain the components present in a typical sensor node. (08 Marks)
- c. Analyze category 1 wireless sensor network applications for :
i) Highway monitoring
ii) Wildfire instrumentation. (06 Marks)

OR

- 6 a. With a neat diagram, explain the basic sensor network architectural elements. (06 Marks)
- b. Analyze category 2 wireless sensor network applications for home control and building automation. (08 Marks)
- c. Explain the challenges and hurdles of wireless sensor networks. (06 Marks)

Module-4

- 7 a. With a neat block diagram, explain the hardware and software components of Wireless Nodes (WNs). (10 Marks)
- b. Explain the following wireless transmission technologies for campus applications :
 i) Bluetooth
 ii) WLAN. (10 Marks)

OR

- 8 a. Explain the wireless node (WN) operating environment with respect to :
 i) Resource constraints
 ii) Design constraints. (10 Marks)
- b. Explain the 3G cellular networks (a wireless transmission technology) for MAN/WAN applications. (10 Marks)

Module-5

- 9 a. Explain the Wireless Sensor Network (WSN) middleware principles. (06 Marks)
- b. Explain the data related functions for middleware in Wireless Sensor Networks (WSN). (08 Marks)
- c. Explain the different performance metrics of Wireless Sensor Networks (WSNs). (06 Marks)

OR

- 10 a. Explain general middleware architecture for wireless sensor networks (WSNs). (06 Marks)
- b. Explain the following related to Wireless Sensor Network (WSN) design issues :
 i) MAC protocols
 ii) Routing protocols
 iii) Transport protocols. (08 Marks)
- c. Explain the basic performance models of wireless sensor networks (WSNs). (06 Marks)

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Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Multimedia Communication

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Describe with the aid of a diagram, how a PSTN can support range of Multimedia Common Applications. (10 Marks)
- b. Classify five types of Communication network that are used to provide multimedia services. (10 Marks)

OR

- 2 a. Discuss the term interactive television with the help of a neat diagram. (10 Marks)
- b. Outline the communication modes available to transfer the information stream. (10 Marks)

Module-2

- 3 a. Describe the function of signal encoder with the associated waveform. (10 Marks)
- b. Assuming the bandwidth of a speech signal is from 50 Hz through to 10 KHz and that of a music signal is from 15 Hz through to 20 KHz, compute the bit rate that is generated by the digitization procedure in each case assuming the Nyquist sampling rate is used with 12 bits per sample for the speech signal and 16 bits per sample for the music signal. Derive the memory required to store a 10 min passage of stereophonic music. (10 Marks)

OR

- 4 a. Outline Raster – Scan operation associated waveform. (10 Marks)
- b. Calculate the time to transmit the following digitized images at both 64 Kbps and 1.5 Mbps :
 - i) A $640 \times 480 \times 8$ VGA – Compatible image.
 - ii) A $1024 \times 768 \times 24$ SVGA – Compatible image. (10 Marks)

Module-3

- 5 a. A series of messages is to be transferred between Two Computers over a PSTN. The messages comprises just the characters A through H. Analysis has shown that the probability (relative frequency of occurrence) of each character is as follows :
A and B = 0.25, C and D = 0.14, E, F, G and H = 0.055.
 - i) Use Shannon's formula to calculate the minimum average number of bits per character.
 - ii) Use Huffmann coding to compute a code work set and show that this is the minimum set by constructing the corresponding Huffmann code tree. (10 Marks)
- b. Describe JPEG Encoder with the aid of diagrams. (10 Marks)

OR

- 6 a. Derive the code for the string "went". Comprising characters with probability of e = 0.3, n = 0.3, t = 0.2, w = 0.1, • = 0.1, using Arithmetic coding. (10 Marks)
- b. Discuss GIF and TIFF format. (10 Marks)

Module-4

- 7 a. Describe DPCM encoder and decoder with a neat diagram. (10 Marks)
b. A digitized video is to be compressed using the MPEG – I standard. Assuming a frame sequence of 1 B B P B B P B B P B B 1..... and average compression ratio of 10:1(I) , 20 : 1(P) and 50:1(B), calculate the average bit rate that is generated by encoder for both the NTSC and PAL digitization formats. (10 Marks)

OR

- 8 a. Describe linear predictive coding encoder and decoder with neat schematic. (10 Marks)
b. Illustrate H-261 Video encoder principles with a necessary diagram. (10 Marks)

Module-5

- 9 a. Discuss the frame format and operational parameters of Ethernet / IEEE 802.3. (10 Marks)
b. Describe the physical and MAC sub – layer of LAN protocol. (10 Marks)

OR

- 10 a. Compare the LAN protocols and Protocol framework. (10 Marks)
b. Describe in detail with diagrams the token ring configuration , frame formats , frame transmission and reception with priority operation. (10 Marks)

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Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Advanced Mathematical Methods

Time: 3 hrs.

Max. Marks: 100

Note : 1. Answer any FIVE full questions, choosing ONE full question from each module.
 2. Provide Normal distribution, t -distribution and F -distribution tables.

Module-1

- 1 a. Transform the matrix

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 1 & 2 \\ 4 & 2 & 1 \end{bmatrix}$$

into tri-diagonal form by Given's method. Using Strum's sequence ,

obtain exact eigen values or the interval of unit length each containing one eigen value.

(10 Marks)

- b. Perform two iterations of the Bridge – Vieta method to find a real root of the equation $x^4 - 3x^3 + 3x^2 - 3x + 2 = 0$. Use the initial approximation $P_0 = +0.5$ and also find the deflated polynomial. (10 Marks)

OR

- 2 a. Transform the matrix

$$A = \begin{bmatrix} 1 & 3 & 4 \\ 3 & 2 & -1 \\ 4 & 1 & 1 \end{bmatrix}$$

into tri – diagonal form by Householder's method. Using Strum's

sequence , obtain exact eigen value or the interval of unit length each containing one eigen value.

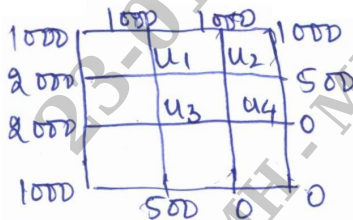
(10 Marks)

- b. Perform two iterations of the Bairstow method to extract a quadratic factor $x^2 + px + q$ from the polynomial $x^3 + x^2 - x + 2 = 0$. Use the initial approximations $P_0 = -0.9$, $q_0 = 0.9$.

(10 Marks)

Module-2

- 3 a. Solve the Laplace equation
- $U_{xx} + U_{yy} = 0$
- for the square mesh of the below figure with boundary values as given below :



Carryout three iterations.

(10 Marks)

- b. Solve the Poisson's equation $U_{xx} + U_{yy} = -81xy$. Where $0 < x < 3$, $0 < y < 3$ given that $u(0, y) = 0$, $u(x, 0) = 0$, $u(3, y) = 100$, $u(x, 3) = 100$ and $h = 1$. (10 Marks)

OR

- 4 a. Use the Crank – Nicolson scheme to solve the heat equation problem

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} \text{ with the conditions : } u(x, 0) = \sin x, \quad 0 \leq x \leq \pi, \quad u(0, t) = u(\pi, t), \quad t > 0.$$

(10 Marks)

- b. Obtain the solution of $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$ by taking $h = 0.2$, where $0 \leq x \leq 1$, subjected to the initial conditions $u(x, 0) = \sin \pi x$, $u_t(x, 0) = \sin \pi x$ and the boundary conditions $u(0, t) = 0$, $u(1, t) = 0$ by using explicit method.

(10 Marks)

Module-3

- 5 a. Find the Taylor's expansion of $f(z) = \frac{2z^3 + 1}{z^2 + 1}$ in the region $z = i$. (06 Marks)

- b. Evaluate $\oint \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2(z-2)} dz$, where C is the circle $|z| = 3$. (07 Marks)

- c. Show that $\int_0^{2\pi} \frac{\cos 2\theta}{(1-2a \cos \theta + a^2)} d\theta = \frac{2\pi a^2}{(1-a^2)}$, $a^2 < 1$ by using contour integration. (07 Marks)

OR

- 6 a. Find the Laurent's expansion of $f(z) = \frac{7z-2}{(z-1)(z-2)}$ about the point $1 < z+1 < 3$. (06 Marks)

- b. Evaluate $\oint_C \frac{z-3}{(z^2+2z+5)} dz$, where C is the circle $|z+1-i| = 2$. (07 Marks)

- c. Evaluate $\int_0^\infty \frac{\cos ax}{x^2+1} dx$ by using Contour integration. (07 Marks)

Module-4

- 7 a. Two independent sample of eight and seven items respectively had the following values of the variable :

Sample 1 :	9	11	13	11	15	9	12	14
Sample 2 :	10	12	10	14	9	8	10	

Do the two estimates of population variance differ significantly at 5% level of significance ($F_{0.05} = 4.21$). (10 Marks)

- b. Set up an analysis of variance (ANOVA) table to assess the significance of possible variation in performance in a certain test between the schools of a city. A common test was given to a numbers of students taken at random from the fifth class of the three schools concerned the results given below :

A	B	C
9	13	14
11	12	13
13	10	17
9	15	7
8	5	9

Construct the analysis of variance for the given data. (10 Marks)

OR

- 8 a. The mean and S.D of the maximum loads supported by 60 cables are 11.09 tonnes and 0.73 tonnes respectively. Find i) 95% ii) 99% confidence limits for mean of the maximum loads of all cables produced by the company. (10 Marks)
- b. Measurements on the length of a copper were taken by two experiments A & B are as under

A's measurements	12.29	12.25	11.86	12.13	12.44	12.78	12.77	11.90	12.47
B's measurements	12.39	12.46	12.34	12.22	11.98	12.46	12.23	12.06	

Test whether B's measurements are more accurate than A's at 5% level of significance.

(10 Marks)

Module-5

- 9 a. Explain i) Absorbing state ii) Transient state iii) Recurrence state. (06 Marks)
- b. Three boys A, B and C are throwing a ball to each other. A always throws the ball to B and B always throws the ball to C. But C is just as likely to throw the ball to B as to A. If C was the first person to throw the ball, find the probability that A has the ball for the forth throw. (07 Marks)
- c. Customers arrive at a one – man barber shop according to a Poisson process with a mean inter arrival time of 12 min. Customers spend an average of 10mm in the barber's chair
- i) What is the expected number of customers in the barber shop and in queue?
- ii) Calculate the percentage of time an arrival can walk straight into barber's chair without having to wait. (07 Marks)

OR

- 10 a. Find the unique fixed probability vector for the regular stochastic matrix.

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

(06 Marks)

- b. A student's study habits are as follows. If he studies one night , he is 70% sure not to study the next night. On the other hand , if he does not study one night , he is 60% sure not to study the next night. In the long run how often does he study? (07 Marks)
- c. Given an average arrival rate of 20 per hour, is it better for a customer to get service at a single channel with mean service rate of 22 customers per hour or at one of two channels is parallel with mean service rat at 11 customers per hour for each of the two channels. Assume both the queues to be Poisson type. (07 Marks)

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Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Research Methodology and Intellectual Property Rights

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Give various definitions of research explaining its meaning and scope. Explain the objective and motivation in Engineering research. (08 Marks)
- b. Describe the steps involved in research process with neat diagram. (06 Marks)
- c. What is ethics in Engineering research? Why it is important. (06 Marks)

OR

- 2 a. What is research problem? Define the main issues which should receive the attention of the research in formulating the research problem. (08 Marks)
- b. Discuss in detail various types of research. (06 Marks)
- c. Describe different types of research misconduct. (06 Marks)

Module-2

- 3 a. Discuss the importance of critical literature review and its uses in planning innovation research. (06 Marks)
- b. Explain the concept of knowledge flow through citation. How collaborations certainly impact citation counts? Explain. (08 Marks)
- c. Write short notes on : (06 Marks)
 - i) Google and Google scholar
 - ii) Acknowledgment and Attributions

OR

- 4 a. Analyze the following terms which do not fulfill the actual goal of citations and acknowledgements. i) Spurious citations ii) Biased citations iii) Self citations iv) Coercive citations. (08 Marks)
- b. What are the things author should acknowledge? (06 Marks)
- c. Elaborate on the following : (06 Marks)
 - i) Technical reading
 - ii) Critical and creative reading.

Module-3

- 5 a. What are Intellectual Property Rights? Explain the necessity of it. (06 Marks)
- b. Differentiate between invention and Innovation. What is a Patent? What are the criteria of patentability? (08 Marks)
- c. Enumerate the procedure for application preparation filing and grant of patents. (06 Marks)

OR

- 6 a. List and explain in detail about various types of Intellectual property rights. (08 Marks)
- b. Explain in detail about the infringement of patents. (06 Marks)
- c. What types of invention are not patentable in India? (06 Marks)

Module-4

- 7 a. What is Copyright? Explain classes, criteria and ownership of copyright. (08 Marks)
 b. Discuss about the copyright ownership issues. (06 Marks)
 c. Describe the process involved in the registration of a trademark. (06 Marks)

OR

- 8 a. Enumerate the procedure for registration of copyright. (08 Marks)
 b. Explain about the Trademark and rights from trademarks registration. (06 Marks)
 c. Explain about types of Trademarks registered in India. (06 Marks)

Module-5

- 9 a. How can industrial designs be protected? (06 Marks)
 b. What is a geographical indication? Explain the following with respect to GI.
 i) Ownership and right granted to the GI holders
 ii) Registered GI in India (08 Marks)
 c. Explain the following :
 i) Famous Industrial Designs
 ii) Generic GI and Homonymous GI (06 Marks)

OR

- 10 a. Explain the classification of Industrial Designs. (06 Marks)
 b. Explain the procedure for GI registration. (08 Marks)
 c. Explain about the enforcement of IPR in India. (06 Marks)

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Third Semester B.E. Degree Examination, Dec.2024/Jan.2025 Network Theory

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define the following terms : (i) Network and Circuit (ii) Active and Passive circuits (iii) Mesh and Loop (iv) Node (06 Marks)
- b. Reduce the circuit using source transformation technique for the circuit shown in Fig. Q1 (b). (07 Marks)

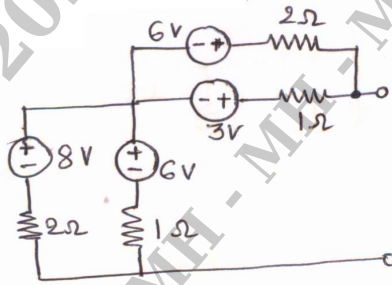


Fig. Q1 (b)

- c. Find the voltage across capacitor using mesh analysis for the circuit shown in Fig. Q1 (c). (07 Marks)

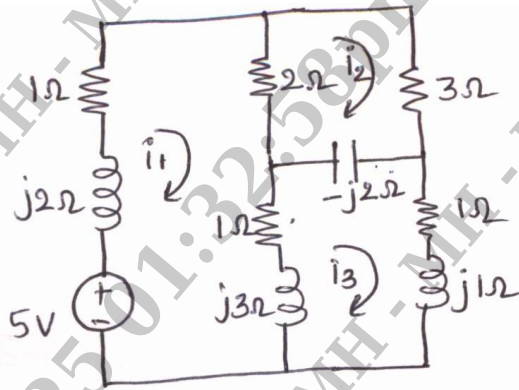


Fig. Q1 (c)

OR

- 2 a. Derive the expression for, (i) Δ to Y transformation (ii) Y to Δ transformation , (10 Marks)
 Δ - delta, Y - Star
- b. Use the nodal analysis to find the value of V_x in the circuit shown in Fig. Q2 (b), such that the current through $(2 + j3)\Omega$ is zero. (10 Marks)

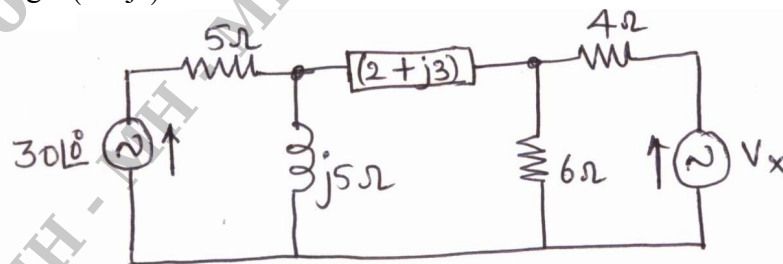


Fig. Q2 (b)

Module-2

- 3 a. State superposition theorem and hence obtain the response I for the network shown in Fig. Q3 (a).

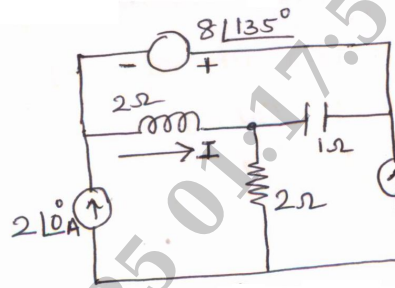


Fig. Q3 (a)

(12 Marks)

- b. State and prove Millman's theorem.

(08 Marks)

OR

- 4 a. Obtain the Thevenin's and Norton's Equivalent circuit for the network shown in Fig. Q4 (a).

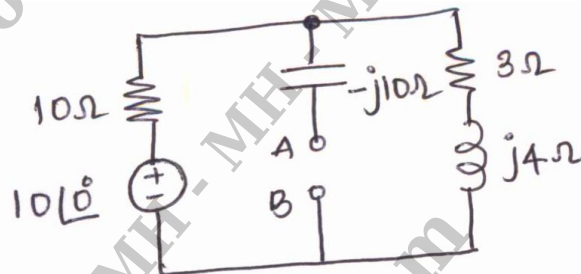


Fig. Q4 (a)

(08 Marks)

- b. State and prove Maximum power transfer theorem for AC circuits. (06 Marks)
 c. Determine the Norton's Equivalent circuit across AB terminals and hence find current through 5Ω resistor for the network shown in Fig. Q4 (c). (06 Marks)

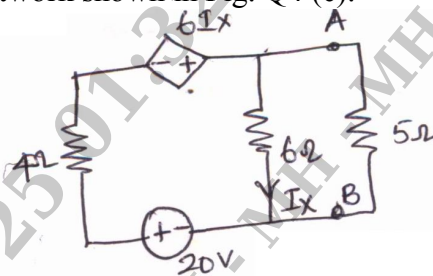


Fig. Q4 (c)

Module-3

- 5 a. Explain in detail the transient response of a R-L circuit for D.C. Excitation. (10 Marks)
 b. Find $i(t)$ for the network shown in Fig. Q5 (b), if the switch K is opened at $t = 0$, assume circuit was in steady state condition before the switch was opened at $t = 0$. (10 Marks)

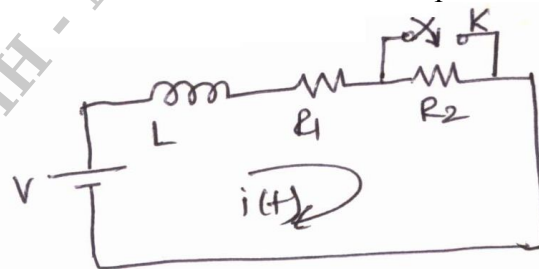


Fig. Q5 (b)

OR

- 6 a. Explain in detail the transient response of R-C circuit for A.C. Excitation. (10 Marks)
- b. Determine $i(0^+)$, $\frac{di(0^+)}{dt}$, $\frac{d^2i(0^+)}{dt^2}$ if switch K is closed at $t = 0$ for the circuit shown in Fig. Q6 (b). Given $V = 10 \text{ V}$, $R = 10 \Omega$, $L = 1 \text{ H}$, $C = 10 \mu\text{F}$ and $V_C = 0$. (10 Marks)

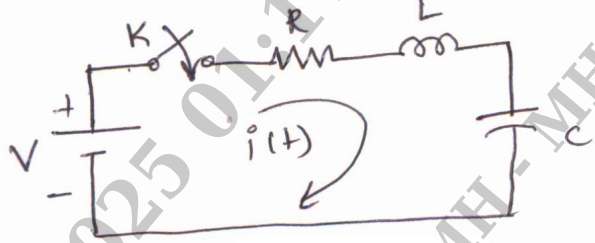


Fig. Q6 (b)

Module-4

- 7 a. State and prove initial value and final value theorem. (10 Marks)
- b. Find the Laplace transform of waveform shown in Fig. Q7 (b). (10 Marks)

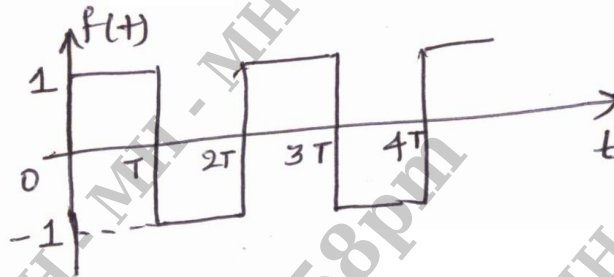


Fig. Q7 (b)

OR

- 8 a. Obtain the Laplace transform of, (i) Step (ii) Ramp (iii) $\sin \omega t$ (10 Marks)
- b. Find the Equivalent impedance of the circuit shown in Fig. Q8 (b), using Laplace transform.

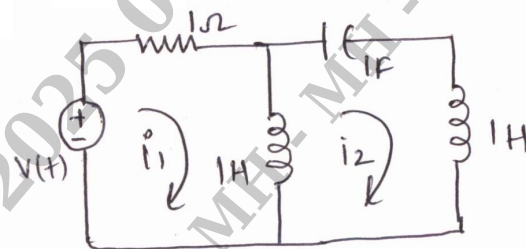


Fig. Q8 (b)

(10 Marks)

Module-5

- 9 a. Define resonant frequency, half power frequencies with respect to series circuit and hence show that resonant frequency is geometric mean of half power frequencies. (08 Marks)
- b. A coil of inductance 10 H and 10Ω resistance is connected in parallel with 100 PF capacitor. The combination is applied with a voltage of 100 V . Find resonant frequency and current at resonance. (06 Marks)
- c. A series RLC circuit consists of a resistance of $1 \text{ K}\Omega$ and an inductance of 100 mH in series with capacitor of 10 PF . If 100 V is applied as input across the combination determine resonant frequency, Max-current, Q-factor. (06 Marks)

OR

- 10 a. Explain z and y parameters with equivalent circuit and also express z-parameters in terms of y-parameters. (08 Marks)
- b. Find the transmission parameters for the circuit shown in Fig. Q10 (b).

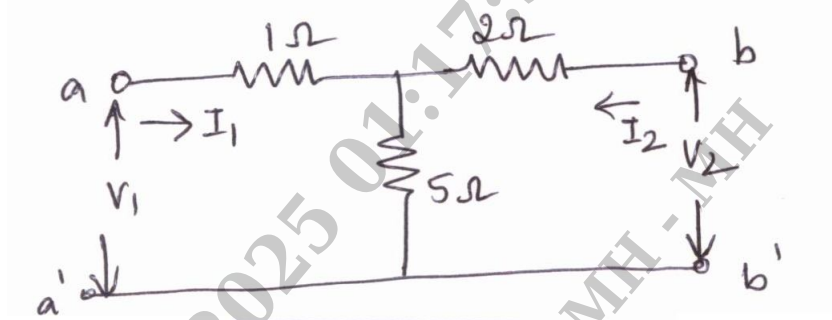


Fig. Q10 (b)

(06 Marks)

- c. Define Q-factor, Selectivity, Bandwidth and hence derive the relation between resonant frequency and half power frequencies. (06 Marks)

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Third Semester B.E. Degree Examination, Dec.2024/Jan.2025 Electronic Devices

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Illustrate the different types of bonding forces in solids with neat diagram. (10 Marks)
- b. Differentiate between Direct and Indirect semiconductor. (10 Marks)

OR

- 2 a. Explain Electron – Hole pair concept with the help of neat diagram and equations. (10 Marks)
- b. What is Hall – Effect? With neat diagram, explain how does hall effect works. (10 Marks)

Module-2

- 3 a. Explain the qualitative description of current flow at P-N junction under equilibrium and biased condition. (10 Marks)
- b. Explain Zener breakdown and avalanche breakdown under reverse biased P-N junction with neat diagram. (10 Marks)

OR

- 4 a. Discuss the operation of light emitting diode with neat sketch. (10 Marks)
- b. How does photodiode works as a photovoltaic cell explain with the help of diagram? (10 Marks)

Module-3

- 5 a. Explain the amplification of a p-n-p transistor with neat diagram. (10 Marks)
- b. Explain the switching operation of a transistor. Also explain the four operating regimes of transistors. (10 Marks)

OR

- 6 a. Derive an expression for I_c and I_E using Ebers Moll equation and draw its equivalent circuit. (10 Marks)
- b. Discuss the fabrication process of BJT. (10 Marks)

Module-4

- 7 a. Discuss the Basic n channel pn JFET operation with equivalent circuit and frequency limitation. (10 Marks)
- b. Explain Ideal capacitance – voltage characteristics of MOSFET. (10 Marks)

OR

- 8 a. Explain two – terminal MOS structure using energy band diagram. (10 Marks)
- b. Explain the principle of operation of p-channel enhancement mode MOSFET with neat diagram and equations. (10 Marks)

Module-5

- 9 a. With Schematic diagram, explain ION – implantation system. (08 Marks)
- b. Explain low pressure chemical vapour deposition reactor. (06 Marks)
- c. Discuss photolithography neat sketch. (06 Marks)

OR

- 10 a. What are the different types of Integrated circuits and its significance? (10 Marks)
- b. Explain the process of Integration involved in CMOS technology. (10 Marks)

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Third Semester B.E. Degree Examination, Dec.2024/Jan.2025 Digital System Design

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With the help of block diagram, explain the general logic design sequence. (06 Marks)
- b. Place the following Boolean equations in to respective canonical forms
 i) $f(abc) = \bar{a}\bar{b} + ac + \bar{b}c$ ii) $f(pqr) = (p + \bar{q})(\bar{q} + r)$ (06 Marks)
- c. Find the minimal sum for the function using K-map $f(abcd) = \sum m(1, 2, 6, 7, 9, 11, 12, 15)$. (08 Marks)

OR

- 2 a. Design a digital circuit for implementing majority of four digital inputs. (07 Marks)
- b. Simplify the given Boolean expression using K-map
 $f(abcd) = \Pi m(1, 2, 3, 4, 9, 10) + \Pi d(0, 14, 15)$ (05 Marks)
- c. Find the prime implicants of the function using Quine-Muclusley method.
 $f(abcd) = \sum m(0, 1, 2, 3, 6, 7, 8, 9, 14, 15)$. (08 Marks)

Module-2

- 3 a. Design the full subtractor circuit using universal gates. (10 Marks)
- b. Design 4:16 decoder using two 3:8 decoder. (10 Marks)

OR

- 4 a. Explain the structure of Programmable Logic Array (PLA) with example. (06 Marks)
- b. Implement $f(pqrs) = \sum m(0, 1, 5, 6, 7, 9, 10, 15)$ using 4:1 MUX with p,q as select lines. (08 Marks)
- c. Implement the following POS function $f(pqr) = \Pi(0, 1, 3, 5)$ using 3:8 decoder with active-high outputs. (06 Marks)

Module-3

- 5 a. What is race around condition? How can it be overcome by MS-JK flip-flop? (08 Marks)
- b. Define register and explain four bit SISO, SIPO, PISO, PIPO registers. (12 Marks)

OR

- 6 a. Bring out differences between synchronous and a synchronous counters. (05 Marks)
- b. With the help of output waveforms, explain 3-bit asynchronous counter. (08 Marks)
- c. Explain the working of twisted ring counter with necessary logic diagram and waveforms. (07 Marks)

Module-4

- 7 a. Design synchronous MOD-7 counter to count from 0 to 6. (12 Marks)
- b. Sequential circuit has one input and one output. The state diagram of the same is shown in Fig Q7(b). Design sequential circuit using D-flip-flop.

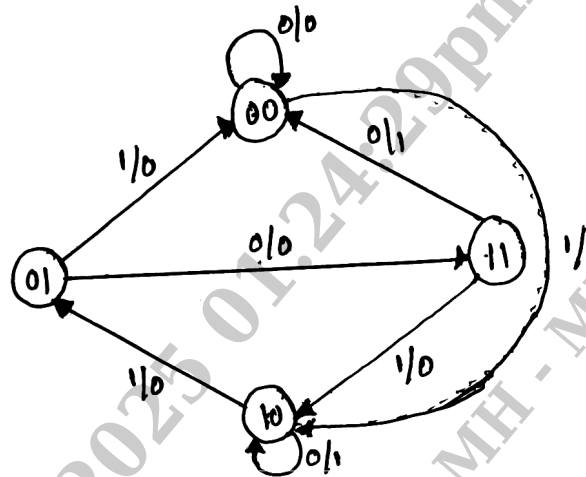


Fig Q7(b)

(08 Marks)

OR

- 8 a. Construct the transition table, state table, state diagram for Moore sequential circuit given in Fig Q8(a)

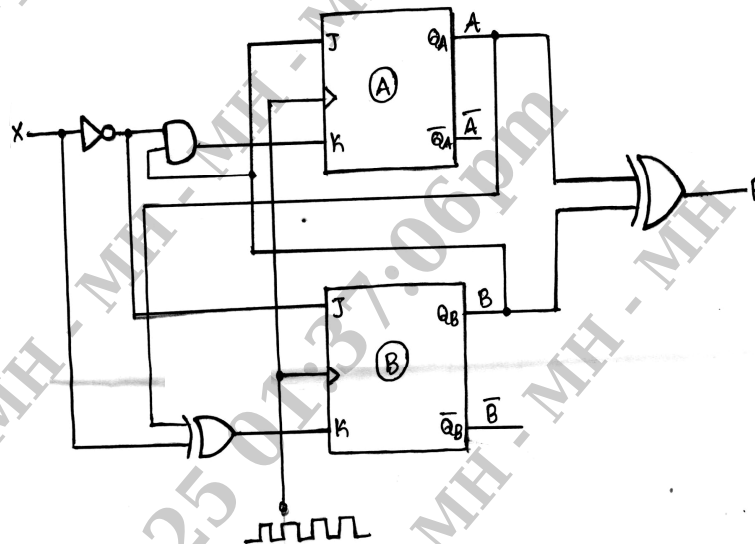


Fig Q8(a)

(10 Marks)

- b. Design a MOD – 6 synchronous counter using D-flipflop for the sequence 0 – 4 – 3 – 5 – 1 – 2

(10 Marks)

Module-5

- 9 a. Design a comparator using iterative circuits. (10 Marks)
b. With neat block diagram, explain serial adder with accumulator. (10 Marks)

OR

- 10 a. Design a binary divider circuits. (10 Marks)
b. List the guide line for constructions of state graphs. (10 Marks)

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18EC52

Fifth Semester B.E. Degree Examination, Dec.2024/Jan.2025 Digital Signal Processing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Compute N-point DFT of the following signals :
i) $x(n) = a^n, 0 \leq n \leq N-1$
ii) $x(n) = 1, 0 \leq n \leq N-1$. (10 Marks)
- b. Determine 4-point circular convolution of the sequences.
 $x_1(n) = \{2, 1, 2, 1\}$ and $x_2(n) = \{1, 2, 3, 4\}$ using graphical method. (05 Marks)
- c. Compute the DFT of the sequence defined by $x(n) = (-1)^n$ for i) $N = 3$ ii) $N = 4$. (05 Marks)

OR

- 2 a. Illustrate the following properties of DFT :
i) Linearly
ii) Circular time shift (10 Marks)
- b. Compute the IDFT of 4-point sequence :
 $X(K) = \{4, -j2, 0, j2\}$ using DFT. (10 Marks)

Module-2

- 3 a. Develop radix - 2 decimation - in - time FFT algorithm and write signal flow graph for $N = 8$. (10 Marks)
- b. i) Compute the 4-point DFT of the sequence $x(n) = \{1, 0, 1, 0\}$ using DIT FFT radix - 2 algorithm.
ii) Find $x(n)$ for $X(K)$ found in part(i) by DIF FFT algorithm. (10 Marks)

OR

- 4 a. Find the o/p $y(n)$ of a filter whose impulse response is $h(n) = \{3, 2, 1, 1\}$ and input $x(n) = \{1, 2, 3, 3, 2, 1, -1, -2, -3, 5, 6, -1, 2, 0, 2, 1\}$ using overlap - add method assuming the length of block is 7. (10 Marks)
- b. Explain overlap-save method to find the output of the filter. (10 Marks)

Module-3

- 5 a. Explain any three window functions to design FIR filters. (10 Marks)
- b. A lowpass filter is to be designed with the following desired frequency response

$$H_d(e^{jw}) = H_d(w) = \begin{cases} e^{-j2w}, & |w| < \frac{\pi}{4} \\ 0, & \frac{\pi}{4} \leq |w| < \pi \end{cases}$$

Determine the filter coefficients $h_d(n)$ and $h(n)$ if $w(n)$ is a rectangular window defined as follows :

$$w_R(n) = \begin{cases} 1, & 0 \leq n \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

Also, find the frequency response, $H(w)$ of the resulting FIR filter.

(10 Marks)

OR

- 6 a. Realize the FIR filter whose impulse response is given by

$$h(n) = \delta(n) + \frac{1}{4}\delta(n-1) - \frac{1}{8}\delta(n-2) + \frac{1}{4}\delta(n+3) + \delta(n-4) .$$
 (10 Marks)
- b. Consider a three stage FIR Lattice structure having the co-efficient $K_1 = -0.65$, $k_2 = -0.34$ and $k_3 = 0.8$. Realize this filter in direct form. (10 Marks)

Module-4

- 7 a. Compare IIR filter with FIR filter. (10 Marks)
- b. Derive an expression for the order of analog Butterworth prototype low pass filter. (10 Marks)

OR

- 8 a. Design an Buterworth filter for which gain $K_p = 0.5$, $K_s = 0.1$ and passband frequency is 2 rad/sec, stopband frequency is 10 rad/sec. (10 Marks)
- b. Draw the block diagrams of direct form – I realizations for a digital IIR filter described by the system function :

$$H(z) = \frac{8z^3 - 4z^2 + 11z - 2}{\left(z - \frac{1}{4}\right)\left(z^2 - z + \frac{1}{2}\right)} .$$
 (10 Marks)

Module-5

- 9 a. Discuss briefly the following DSP hardware units :
 i) MAC unit ii) Shifter iii) Address generators. (10 Marks)
- b. Convert the following decimal numbers into Q – 15 representation :
 i) 0.560123 ii) 0.160123. (10 Marks)

OR

- 10 a. Explain briefly the basic architecture of TMS320C54X family processor. (10 Marks)
- b. Discuss the following IEEE floating – point formats
 i) Single precision format
 ii) Double precision format. (10 Marks)

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18EC53

Fifth Semester B.E. Degree Examination, Dec.2024/Jan.2025 Principles of Communication Systems

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the generation of amplitude modulated (AM) waves using switching modulator. (08 Marks)
- b. With a neat block diagram, explain the working of COSTAS receiver used for demodulation of DSB-SC singles. (07 Marks)
- c. An audio signal $10 \sin 1000 \pi t$ volts is used to amplitude modulate a carrier signal $75 \sin (2\pi \times 10^6)t$. Assume modulation index as 0.5. Find :
- Side band frequencies
 - Amplitude of each side band
 - The bandwidth required
 - The total power delivered to a load of 100Ω . (05 Marks)

OR

- 2 a. With a neat block diagram and equations, explain the generation and demodulation of VSB signals. (08 Marks)
- b. Fig.Q2(b) shows the block of an AM system with $s(t) = AC[1 + K_a m(t)]\cos(2\pi f_c t)$ and $|K_a m(t)| < 1$ for all t . If $m(t)$ is a band limited signal in the interval $(-w < f < w)$ and $f_c > 2w$ show that $m(t)$ can be obtained from the square rooter output.

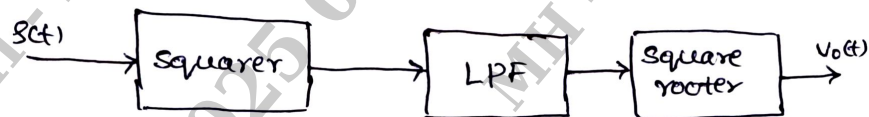


Fig.Q2(b)

- c. Explain in detail the scheme of FDM. (07 Marks) (05 Marks)

Module-2

- 3 a. Derive the equation of FM wave and list the properties of angle modulated waves. (07 Marks)
- b. Obtain the time domain expression of NBFM wave plot its spectrum and compare with AM using phasor diagrams. (08 Marks)
- c. An angle modulated wave is defined by the equation :
 $s(t) = 10 \cos[2\pi \times 10^6 t + 5 \sin(2000\pi t) + 10 \sin(3000\pi t)]$
determine the following :
- Power in the modulated signal across a standard 1Ω resistor
 - Frequency deviation
 - The deviation ratio
 - Phase deviation
 - Transmission bandwidth. (05 Marks)

OR

- 4 a. With a neat block diagram and necessary equations explain the demodulation of FM waves using non-linear model of phase locked loop (PLL). (08 Marks)
- b. Draw the block diagram of super heterodyne receiver from AM reception and explain the functions of each block. (06 Marks)

- c. An FM signal $s(t) = A_c \cos \left[2\pi f_c t + 2\pi k_f \int_0^t m(t) dt \right]$ is applied to a system consisting of RC

high pass filter and envelope detector shown in Fig.Q4(c). Assume $R \ll X_C$ and envelope detector does not load the filter, determine the envelope detector output assuming $k_f |m(t)| < f_c$ for all t . Comment on the output.

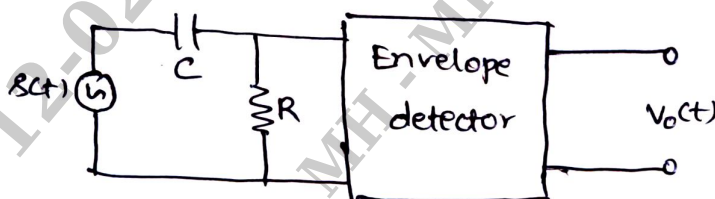


Fig.Q4(c)

(06 Marks)

Module-3

- 5 a. Explain thermal noise and white noise. (06 Marks)
- b. Define noise equivalent bandwidth and derive the expression for the same. (08 Marks)
- c. An amplifier operating over a frequency range of 450 to 460 KHz is having an input resistance of 10 KΩ. If the temperature is 15°C. Find :
- rms noise voltage at the input of the amplifier
 - the amplifier noise power
 - the power spectral density.

(06 Marks)

OR

- 6 a. Starting from fundamentals, derive the expression for Figure of Merit (FOM) of an AM receiver operating on single tone modulation. (10 Marks)
- b. Discuss the noise in FM receivers and obtain the expression for Figure of Merit (FOM) for FM receiver. (10 Marks)

Module-4

- 7 a. Mention the advantages of digitalizing analog signals. (04 Marks)
- b. With relevant equations, state and explain sampling theorem for low pass signals and derive the interpolation formula. (10 Marks)
- c. With a neat block diagram, explain the generation of Pulse Position Modulation (PPM) signals. (06 Marks)

OR

- 8 a. Mention the few applications of pulse amplitude modulation (PAM). (04 Marks)
- b. Consider the signal $x(t) = 5\cos(2000\pi t) + 10\cos(6000\pi t)$
- What is the Nyquist rate and Nyquist interval
 - Assume if the signal is sampled at frequency $f_s = 5000$ Hz ; what is the resulting signal
 - Draw the spectrum of the sampled signal for $f_s = 5000$ Hz. (10 Marks)
- c. With a neat block diagram, explain Time Division Multiplexing Technique(TDM).(06 Marks)

Module-5

- 9 a. A PCM system uses uniform quantizer followed by a N bit encoder. Show that rms signal to quantization noise is approximately given by $[SNR]_{\text{dB}} = (4.8 + 6N)\text{dB}$. (08 Marks)
- b. Explain the generation and reconstruction of PCM signal. (06 Marks)
- c. A TV Signal with a bandwidth of 4.2MHz is transmitted using binary PCM. The number of representation levels are 512 calculate : (06 Marks)
- i) Code word length
 - ii) Final bit rate
 - iii) Transmission band width.

OR

- 10 a. For the bit sequence 10011101 draw unipolar NRZ, polar NRZ, unipolar RZ, bipolar RZ and Manchester encoding formats. (08 Marks)
- b. Explain Delta modulation with relevant equations and waveforms. (06 Marks)
- c. With a neat block diagram, explain the working of linear predictive vocoder. (06 Marks)

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18EC54

Fifth Semester B.E. Degree Examination, Dec.2024/Jan.2025 Information Theory and Coding

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Discuss the reasons for using logarithmic measure of measuring the amount of information. (06 Marks)
- b. A source transmits two independent messages with probabilities of p and $(1-p)$ respectively. Prove that the entropy is maximum when both the messages are equally likely. Plot the variations of entropy (H) as a function of probability ' p ' of the messages. (04 Marks)
- c. Find G_1 and G_2 and verify that $G_1 > G_2 > H(s)$ for the Fig.Q1(c). (10 Marks)

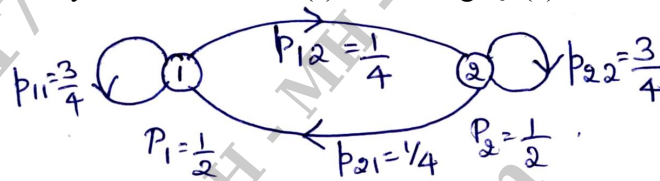


Fig.Q1(c)

OR

- 2 a. Define the following with respect to information theory :
i) Self information
ii) Entropy
iii) Rate of information. (06 Marks)
- b. An analog signal is band limited to 500 Hz and is sampled at "Nyquist rate". The samples are quantized into 4 levels and each level represent one message. The quantization levels are assumed to be independent. The probabilities of occurrence of 4 levels are $P_1 = P_4 = \frac{1}{8}$ and $P_2 = P_3 = \frac{3}{8}$ find the information rate of the source. (04 Marks)
- c. The state diagram of the Mark off source is as shown in the Fig.Q2(c). Find :
i) The entropy of each state H_i
ii) The entropy of source H
iii) G_1, G_2 and $H(G_1 > G_2 > H)$. (10 Marks)

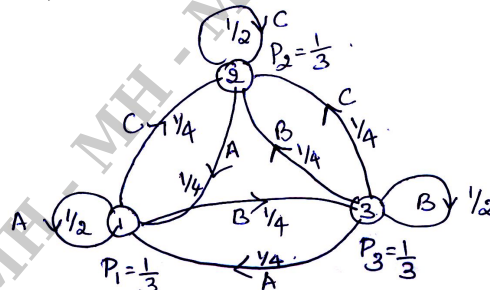


Fig.Q2(c)

Module-2

- 3 a. A DMS has an alphabet $S = \{s_0, s_1, s_2, s_3, s_4, s_5, s_6\}$ and source statistics $P = \{0.125, 0.0625, 0.25, 0.0625, 0.125, 0.125, 0.25\}$. Construct binary Huffman code. Also find the efficiency and redundancy of coding. (10 Marks)
- b. Explain prefix coding with an example. Also explain the properties of prefix codes. (10 Marks)

OR

- 4 a. Explain Shannon's encoding algorithm. State the properties of Shannon's encoding algorithm. (10 Marks)
- b. Apply Shannon – Fano encoding algorithm to the following set of messages and obtain the entropy and efficiency.

Message	m ₁	m ₂	m ₃	m ₄	m ₅	m ₆	m ₇	m ₈
Probability of message	$\frac{16}{32}$	$\frac{4}{32}$	$\frac{4}{32}$	$\frac{2}{32}$	$\frac{2}{32}$	$\frac{2}{32}$	$\frac{1}{32}$	$\frac{1}{32}$

(10 Marks)

Module-3

- 5 a. Prove that the mutual information of the channel is symmetric i.e. $I(X;Y) = (Y;X)$. (08 Marks)
- b. Two noisy channels are cascaded whose channel matrices are given by,

$$p(y_j | x_i) = \begin{bmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \end{bmatrix} \text{ and } p(z_j | y_i) = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & 0 \\ \frac{2}{3} & 0 & \frac{1}{3} \\ 0 & \frac{1}{3} & \frac{2}{3} \end{bmatrix}$$

With $P(x_1) = P(x_2) = 0.5$. Show that $I(X;Y) > I(X;Z)$.

(12 Marks)

OR

- 6 a. State channel capacity theorem : In the channel capacity equation when the signal power is fixed and white Gaussian noise is present, the channel capacity approaches an upper limit with increase in band width 'B'. Prove that this upper limit is given as,

$$C_{\infty} = \lim_{B \rightarrow \infty} C = 1.44 \frac{S}{N_0}$$

(10 Marks)

- b. For the channel shown in Fig.Q6(b) the symbols are transmitted at the rate of 10,000 per second. Calculate maximum mutual information of this channel. (10 Marks)

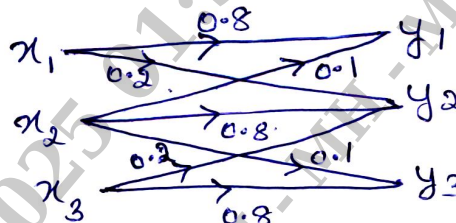


Fig.Q6(b)

Module-4

- 7 a. Consider a (7, 4) linear code whose generator matrix is G

$$G = \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{array} \right]$$

Find :

- All the code vectors of this code
 - Parity check matrix of this code
 - The maximum weight of this code.
- (10 Marks)
- b. The generator polynomial for a (15, 7) cyclic code is $G(x) = 1 + x^4 + x^6 + x^7 + x^8$.
- Find the code vector in systematic form for the message $D(x) = x^2 + x^3 + x^4$
 - Assume that the first and last bit of the code vector $V(x)$ for $D(x) = x^2 + x^3 + x^4$ suffer transmission errors. Find the syndrome of $V(x)$.
- (10 Marks)

OR

- 8 a. For a(5, 2) linear, systematic block code, choose the generator matrix and parity check matrix with the objective of maximizing $d_{\min}$. For the matrix chosen, construct the standard array. (10 Marks)

- b. Consider a(6, 3) linear block code whose

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

- Find all the code vector
- Find all Hamming weight and distance
- Find minimum weight parity check matrix
- Draw encoder circuit for above code.

(10 Marks)

Module-5

- 9 a. Consider (3, 1, 2) convolution encoder with impulse response

$$g_1^{(1)} = \{1 \ 1 \ 0\}, \quad g_1^{(2)} = \{1 \ 0 \ 1\}, \quad g_1^{(3)} = \{1 \ 1 \ 1\}$$

- Draw the encoder block diagram
- Find the generator matrix and output code vector for $m = \{1 \ 1 \ 1 \ 0 \ 1\}$.
- Find the code vector corresponding to the message sequence using time domain approach.

(12 Marks)

- b. Write a note on Viterbi algorithm for decoding of convolutional codes. (08 Marks)

OR

- 10 a. For the convolutional encoder of Fig.Q10(a) determine the following :

- Dimension of the code
- Code rate
- Constraint length
- Generating sequences (impulse responses)
- Output sequence for message sequence of $m = \{1 \ 0 \ 0 \ 1 \ 1\}$ using transfer domain approach.

(08 Marks)

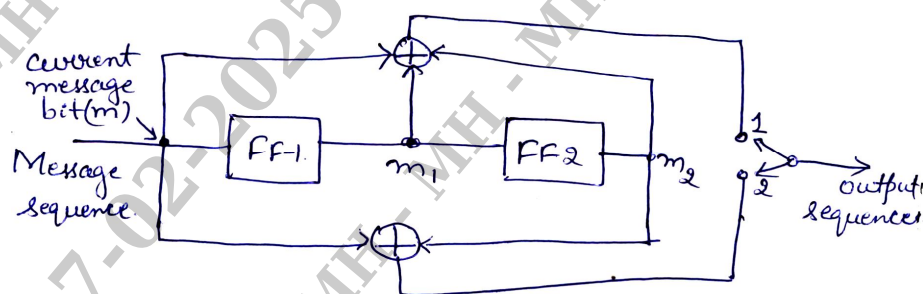


Fig.Q10(a)

- b. A rate 1/3 convolution encoder has generating vectors as :

$$g_1 = (1 \ 0 \ 0), \quad g_2 = (1 \ 1 \ 1), \quad g_3 = (1 \ 0 \ 1)$$

- Sketch the encoder configuration
- State diagram and code tree
- If input message sequence is 10110, determine the output sequence of the encoder using code tree.

(12 Marks)

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18EC55

Fifth Semester B.E. Degree Examination, Dec.2024/Jan.2025 Electromagnetic Wave

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. State and explain Coulomb's law in vector form. (10 Marks)
b. If $\vec{D} = xy^2z^2 \vec{a}_x + x^2yz^2 \vec{a}_y + x^2y^2z \vec{a}_z$ C/m² find i) an expression for ρ_v ii) the total change within the cube defined by $0 \leq x \leq 2$; $0 \leq y \leq 2$; $0 \leq z \leq 2$. (10 Marks)

OR

- 2 a. Obtain an expression for electric field intensity due to infinite line charge. (10 Marks)
b. Define the following terms in electric field density i) Line charge ii) Surface charge iii) volume charge. (10 Marks)

Module-2

- 3 a. State and prove Gauss law for point charge. (05 Marks)
b. State and prove divergence theorem. (05 Marks)
c. Give the electrical tube density $D = 0.3r^2 \vec{a}_r$ nC/m² in free space.
i) Find E at Pt. P($r = 2$; $\theta = 25^\circ$; $\phi = 90^\circ$).
ii) Find the total change within the sphere $r = 3$
iii) Find the total electric flux leaving the sphere $r = 4$. (10 Marks)

OR

- 4 a. Obtain an expression for integral form of work done in moving a Pt. Charge Q from one position to another position. (08 Marks)
b. Calculate the work done in moving a 4C charge from B(1, 0, 0) to A(0, 2, 0) along the path $y = 2 - 2x$, $z = 0$ in the field $E = (1) 5 \vec{a}_x$ V/M (2) $5x \vec{a}_x$ V/m (06 Marks)
c. A 15 nC point charges ρ_s at the origin in free space. Calculate V_1 if point P is located at P(-2, 3, -1) and $V = 0$ at (6, 5, 4). (06 Marks)

Module-3

- 5 a. Drive the Poisson's and Laplaces equations. (08 Marks)
b. State the prove the Stoke's theorem. (06 Marks)
c. Let $V = 2xy^2z^3$ and $E = E_0$ given point P(1, 2, -1). Calculate i) V at P ii) E at P iii) ρ_v at P. (06 Marks)

OR

- 6 a. State and prove the Amperes circuital law. (06 Marks)
b. Drive the expression for vector magnetic potential. (06 Marks)
c. A current element $I d\vec{L} = 10^{-3}(2 \vec{a}_x + 4 \vec{a}_y - \vec{a}_z)$ A/m located at A(-5, 3 -2) produces a field dH at B(3, -4, 3) i) Give a unit vector in the direction at dH at B ii) Find d(H) at B. (08 Marks)

Module-4

- 7 a. Derive an expression for the Force between differential current elements in magnetic field. (06 Marks)
- b. The field $B = -2\bar{a}_x + 3\bar{a}_y + 4\bar{a}_z$ mT is present in free space. Find the vector force exerted on a st. wire carrying 12A current in the a_{AB} direction given A(1, 1, 1) and B(2, 1, 1). (08 Marks)
- c. An air core toroid has 500 turns mean radius of 15 cm cross sectional area of 6 cm^2 . The magnetic motive force is 2000 AT. Calculate total reluctance flux, flux density, field intensity inside the core. (06 Marks)

OR

- 8 a. Write note on forces on magnetic materials. (10 Marks)
- b. Write a note on magnetic circuits. (10 Marks)

Module-5

- 9 a. Drive the expression for a stationary closed path in a time varying field statically induced EMF. (06 Marks)
- b. State Maxwell's equation in both point form and in integral form. (06 Marks)
- c. Find the frequency at which conduction current density and displacement current density are equal in a medium with $\sigma = 2 \times 10^{-4}$ and $\epsilon_r = 81$. (08 Marks)

OR

- 10 a. State and explain poynting theorem. (08 Marks)
- b. Define the following terms in uniform plane wave i) phase velocity ii) Intrinsic impedance iii) wave length. (06 Marks)
- c. The depth at penetration in a certain conducting medium is 0.1 m and the frequency of the electromagnetic wave is 1.0 MHz. Find the conductivity of the conducting medium. (06 Marks)

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Third Semester B.E. Degree Examination, Dec.2024/Jan.2025 Transform Calculus, Fourier Series and Numerical Techniques

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Find $L\left\{\frac{\cos at - \cos bt}{t}\right\}$. (06 Marks)
- b. A periodic function of period $\frac{2\pi}{\omega}$ is defined by, $f(t) = \begin{cases} E \sin \omega t, & 0 < t < \frac{\pi}{\omega} \\ 0, & \frac{\pi}{\omega} < t < \frac{2\pi}{\omega} \end{cases}$ (07 Marks)
- c. Solve : $y'' - y' - 2y = 0$; given $y(0) = 0$ and $y'(0) = 6$ by Laplace transformation method. (07 Marks)

OR

- 2 a. Find $L^{-1}\left\{\frac{4s+5}{(s-1)^2(s+2)}\right\}$. (06 Marks)
- b. Apply convolution theorem to evaluate $L^{-1}\left\{\frac{1}{(s^2+1)(s^2+9)}\right\}$. (07 Marks)
- c. Using unit step function, find the Laplaces transform of, $f(t) = \begin{cases} \sin t, & 0 \leq t < \pi \\ \sin 2t, & \pi \leq t \leq 2\pi \\ \sin 3t, & t \geq 2\pi \end{cases}$. (07 Marks)

Module-2

- 3 a. Obtain the Fourier Series for the function, $f(x) = x^2$, $0 < x < 2\pi$. (06 Marks)
- b. Find the Fourier series of $f(x)$,
Where $f(x) = \begin{cases} -\pi, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$. (07 Marks)
- c. Express $f(x) = x$ as a half-range cosine series in $0 < x < 2$. (07 Marks)

OR

- 4 a. Obtain Fourier series for the function, $f(x) = \begin{cases} \pi x, & 0 \leq x \leq 1 \\ \pi(2-x), & 1 \leq x \leq 2 \end{cases}$. (06 Marks)
- b. Find the Fourier half-range cosine series of the function $f(x) = (x+1)$, in $(0, 1)$. (06 Marks)
- c. Compute the first harmonic of the Fourier series of $f(x)$ given in the following table :

x	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{5\pi}{3}$	2π
y	1.0	1.4	1.9	1.7	1.5	1.2	1.0

(08 Marks)

Module-3

- 5 a. Find the z-transform of, $3n - 4\sin\left(\frac{n\pi}{4}\right) + 5a$. (06 Marks)
- b. Compute the inverse z – transform of, $\frac{2z^2 + 3z}{(z+2)(z-4)}$. (07 Marks)
- c. Find the Fourier transform of, $f(x) = \begin{cases} 1, & |x| < 1 \\ 0, & |x| > 1 \end{cases}$, Hence evaluate $\int_0^{\infty} \frac{\sin x}{x} dx$. (07 Marks)

OR

- 6 a. Find the Fourier sine transform of e^{-ax} . (06 Marks)
- b. If $U(z) = \frac{2z^2 + 5z + 14}{(z-1)^4}$, evaluate u_2 and u_3 . (07 Marks)
- c. Using the z-transform, solve $u_{n+2} + 4u_{n+1} + 3u_n = 3^n$, $u_0 = 0$, $u_1 = 1$. (07 Marks)

Module-4

- 7 a. Find by Taylor's series method the value of y at $x = 0.1$ and $x = 0.2$ to four places of decimals from, $\frac{dy}{dx} = x^2y - 1$, $y(0) = 1$. (06 Marks)
- b. Apply Runge-Kutta fourth order method to find an approximate value of y when $x = 0.2$, given $\frac{dy}{dx} = x + y$, $y(0) = 1$. (07 Marks)
- c. If $\frac{dy}{dx} = 2e^x - y$, $y(0) = 2$, $y(0.1) = 2.010$, $y(0.2) = 2.04$ and $y(0.3) = 2.09$, find $y(0.4)$ by employing the Milne's predictor-corrector formula, use corrector formula twice. (07 Marks)

OR

- 8 a. Using modified Euler's method, solve the IVP $\frac{dy}{dx} = x + \sqrt{y}$, $y(0) = 1$ at $x = 0.2$, perform three modifications. (06 Marks)
- b. Using the fourth order Runge-Kutta method, solve the IVP $\frac{dy}{dx} = \frac{1}{x+y}$ at the point $x = 0.5$. Given that $y(0.4) = 1$. (07 Marks)
- c. Given $\frac{dy}{dx} = x^2(1+y)$, $y(1) = 1$, $y(1.1) = 1.233$, $y(1.2) = 1.548$, $y(1.3) = 1.979$. Determine $y(1.4)$ by Adams-Bashforth method. (07 Marks)

Module-5

- 9 a. Using Runge-Kutta method of fourth order solve the differential equation, $\frac{d^2y}{dx^2} = x\left(\frac{dy}{dx}\right)^2 - y^2$, with $y(0) = 1$, $y'(0) = 0$ at $x = 0.2$. (06 Marks)
- b. Derive Euler's equation in the standard form, $\frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) = 0$. (07 Marks)

- c. On which curve the functional,

$$\int_0^{\frac{\pi}{2}} (y'^2 - y^2 + 2xy) dy, y(0) = 0, y\left(\frac{\pi}{2}\right) = 0 \text{ be extremized.}$$

(07 Marks)

OR

- 10 a. Apply Milne's method to compute $y(0.8)$, given that $\frac{d^2y}{dx^2} = 1 - 2y \frac{dy}{dx}$ and

x	0	0.2	0.4	0.6
y	0	0.02	0.0795	0.1762
y'	0	0.1996	0.3937	0.5689

(06 Marks)

- b. Prove that the geodesics on a plane are straight line.

(07 Marks)

- c. Find the extremal of the functional, $I = \int_0^{\frac{\pi}{2}} (y'^2 - y^2 + 4y \cos x) dx$, given that $y(0) = 0$,

$$y\left(\frac{\pi}{2}\right) = 0.$$

(07 Marks)

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Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Advanced Mathematical Methods

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Reduce the matrix $A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 1 & 2 \\ 4 & 2 & 1 \end{bmatrix}$ to tridiagonal form and find its eigen values using

Given's method

(12 Marks)

- b. Find the eigen values of the following matrix by House holder method $A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & 2 \\ -1 & 2 & 1 \end{bmatrix}$

(08 Marks)

OR

- 2 a. Use the Birge Vieta method to find a real root correct upto three decimal places for the equation $x^3 - 11x^2 + 32x - 22 = 0$. Take $P_0 = 0.5$. (12 Marks)
- b. Using 'Bairstow' method, find the quadratic factor of $x^3 + x^2 - x + 2 = 0$ by taking $P = -0.9$, $q = 0.9$. (08 Marks)

Module-2

- 3 a. Solve $u_{xx} + u_{yy} = 0$ for the following square with the boundary conditions as indicated in the Fig Q3(a).

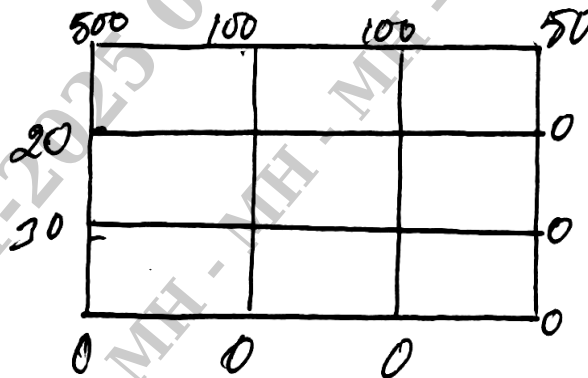


Fig Q3(a)

(10 Marks)

- b. Solve the Poisson equation $u_{xx} + u_{yy} = 8x^2y^2$ for the following square mesh with $u(x,y) = 0$ on the boundary and mesh length equal to 1.

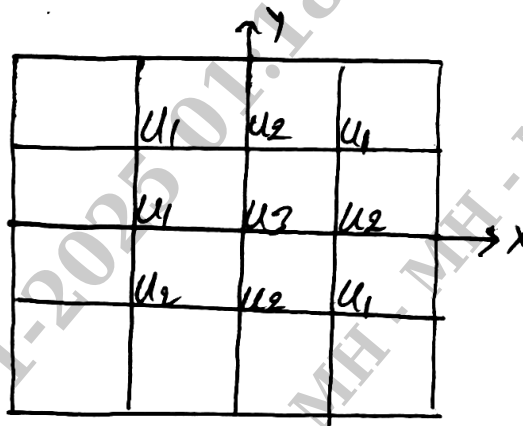


Fig Q3(b)

(10 Marks)

OR

- 4 a. Solve the wave equation $u_{tt} = 4u_{xx}$ subjected to $u(0, t) = 0$, $u(4t) = 0$, $u_t(x, 0) = 0$ and $u(x, 0) = x(4 - x)$ by taking $h = 1$, $k = 0.5$. (10 Marks)
- b. Using Crank – Nicholson method, solve $u_{xx} = 16 u_t$, $0 < x < 1$, $t > 0$, given $u(x, 0) = 0$, $u(0, t) = 50t$. Compute u for two steps in the direction of t by taking $h = 0.25$. (10 Marks)

Module-3

- 5 a. Find the Taylor's series expansion of $f(z) = \frac{1}{z^2 + 4}$ about $z = -i$. (07 Marks)

b. Evaluate $\int_0^{2\pi} \frac{d\theta}{5 - 3\cos\theta}$. (07 Marks)

c. Find the residue of $f(z) = \frac{ze^2}{(z-a)^3}$ at its pole. (06 Marks)

OR

- 6 a. Find the Laurent's series expansion of $f(z) = \frac{1}{(z+1)(z+3)}$ in $1 < |z| < 3$. (07 Marks)

b. Using Cauchy's residue theorem, evaluate $\int_C \frac{(4-3z)dz}{z(z-1)(z-2)}$, where C is the circle $|z| = \frac{3}{2}$. (07 Marks)

c. Evaluate $\int_0^\infty \frac{dx}{(1+x^2)^3}$, using contour integration. (06 Marks)

Module-4

- 7 a. A coin is tossed 900 times and head appears 490 times, can it be concluded that the coin is biased at 5% level of significance. Given $z_{0.05} = 1.96$. (07 Marks)
- b. An examination was conducted to two groups of students of two colleges. The following is the given data :
 College A : $\mu_A = 75, \sigma_A = 9, n_1 = 90$
 College B : $\mu_B = 79, \sigma_B = 7, n_2 = 60$
 Is there a significant difference in the performance of two colleges at 5% level of significance. Given $z_{0.05} = 1.96$. (07 Marks)
- c. The following table represent the production of three varieties of crops A, B, C, each grown on 4 plots. Find whether the variety differences are significant. Given $F_c = 4.26$.

Production/Acre			
Plot	A	B	C
1	6	5	5
2	7	5	4
3	3	3	-3
4	8	7	4

(06 Marks)

OR

- 8 a. A die is thrown 9000 times. The numbers 3 or 4 appears 3240 times. Test whether the die is unbiased at 5% level of significance. Given that $z_{0.05} = 1.96$. (07 Marks)
- b. Two independent samples of sizes 7 and 6 have the following data :
 Sample A : 28 30 32 33 31 29 34
 Sample B : 29 30 30 24 27 28 -
 Examine whether the samples have been drawn from a normal population having the same variance. Given $F_{0.05} = 4.39$. (07 Marks)
- c. Write a short note on :
 i) Null hypothesis ii) Types of errors iii) level of significance. (06 Marks)

Module-5

- 9 a. Show that the process $x(t) = A \cos (wt + \theta)$ is wide sense stationary. Where A and w are constants and θ is uniformly distributed random variable in $(0, 2\pi)$. (07 Marks)

- b. Find the fixed probability vector of the regular Stochastic matrix $A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{pmatrix}$. (07 Marks)
- c. In a railway yard, trains arrive at a rate of 30 trains per day. Assuming that, the inter arrival time and service time distribution are exponential with an average of 36 minutes. Find :
 i) The probability that the yard is empty ii) average queue length, if the line capacity of the yard is 9 trains. (06 Marks)

OR

- 10 a. Three boy A, B and C are throwing a ball to each other. A always throws the ball to B and B always throws the ball to C, but C is just likely to throw the ball to B as to A. If C is the first to throw the ball, find the probabilities that for the fourth throw
i) A has the ball ii) B has the ball iii) C has the ball. (07 Marks)
- b. Cars arrive at a petrol pump, having one petrol unit, in a Poisson Fashion, with an average of 10 cars per hour. The service time is exponentially distributed with a mean of 3 minutes. Find : i) average number of cars in the system ii) average queue length iii) probability that the number of cars in the system is 2. (07 Marks)
- c. The auto correlation function for a stationary process is given by $R_{xx}(\tau) = 25 + \frac{4}{1+6\tau^2}$.
Find the mean and variance of the process $X(t)$. (06 Marks)

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